frontiers in Marine Science

TYPE Original Research
PUBLISHED 10 August 2026
DOI 10.3389/fmars.2026.1883695

Check for updates

OPEN ACCESS

EDITED BY
Lin Liu,
Ministry of Natural Resources, China

REVIEWED BY
Hanhao Zhu,
Zhejiang Ocean University, China
Hao Wang,
Laoshan National Laboratory, China

*CORRESPONDENCE
Jiahua Zhu
zhujiahua1019@hotmail.com

RECEIVED 17 May 2026
REVISED 09 July 2026
ACCEPTED 15 July 2026
PUBLISHED 10 August 2026

CITATION
Liu D, Chen H, Chen J, Ding Y and Zhu J (2026) Performance evaluation of YOLO models for target detection from ocean sidescan sonar imagery.
Front. Mar. Sci. 13:1883695.
doi: 10.3389/fmars.2026.1883695

COPYRIGHT
© 2026 Liu, Chen, Chen, Ding and Zhu. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Performance evaluation of YOLO models for target detection from ocean sidescan sonar imagery

Dandan Liu[1], Hao Chen[1], Jiajie Chen[1], Yuxiang Ding[1] and Jiahua Zhu[2]*

[1]Yancheng Institute of Technology, School of Electrical Engineering, Yancheng, China, [2]National University of Defense Technology College of Meteorology and Oceanology, Changsha, China

Sidescan sonar images suffer from strong seabed clutter, low gray contrast of underwater targets and frequent missed small targets. Current detection models struggle to balance detection accuracy, inference speed and embedded deployment. To solve these problems, this paper selects six mainstream one-stage detection models, YOLOv4, YOLOv6, YOLOv7, YOLOv9, YOLOv13n and YOLO26n, for comparative experiments. Two sidescan sonar datasets D1 and D2 with distinct imaging features are built. Seven metrics are used to quantitatively evaluate overall model performance: Precision, Recall, mAP@0.5, mAP50–95, FPS, FLOPs and parameter count. Experimental results show that imaging quality directly affects detection performance. The traditional heavy model YOLOv4 has numerous parameters and high computational cost. It produces many missed and false detections under heavy noise and fails to support real-time detection on underwater equipment. Lightweight YOLOv13n and YOLO26n show stronger anti-interference ability on noisy sonar images. Among all models, YOLOv6 on D1 achieves the highest overall mAP@0.5 of 0.944, YOLOv9 on D2 delivers the best localization accuracy across multiple IoU thresholds, and YOLO26n reaches an ultra-high inference speed of 526.32 FPS with only 2.375M parameters and 5.2G FLOPs. By comparing performance gaps between the two datasets, we find that parameter size is positively correlated with overfitting risk. Large models easily memorize unique noise textures from training data and have weak generalization. Restricted by limited parameter capacity, lightweight networks naturally suppress overfitting and achieve more stable detection results across different sea conditions. This paper clarify application scenarios for each model based on accuracy, real-time performance, lightweight degree and generalization. YOLOv4 is only suitable for offline analysis in laboratories. YOLOv9 fits high-precision contour mapping tasks such as underwater archaeology. YOLOv13n works for large-area scanning with medium and small underwater vehicles. YOLO26n has the best overall engineering performance. It can be deployed on low-computing devices including portable sonars and miniature underwater vehicles for long-term continuous real-time underwater detection. This study clarifies the performance trade-offs of various YOLO models for underwater target detection on side-scan sonar images. It provides references for model selection of underwater detection systems with different hardware and task requirements. Future work will expand sonar datasets covering multiple sea areas and adopt noise augmentation strategies to further boost generalization and small target detection performance in complex deep-sea environments.

KEYWORDS
overfitting, sidescan sonar, small-sample learning, underwater target detection, YOLO

1 Introduction

The sidescan sonar (SSS)(Chen et al., 2026; Lin et al., 2026; L. Liu et al., 2026; Qiao et al., 2026) and synthetic aperture sonar (SAS) (Zhang et al., 2019b, 2026, 2026) are key technologies in the field of underwater target detection (Chen et al., 2023; Khan et al., 2024; K. Liu et al., 2023). They feature ultra-high image resolution, strong underwater penetration and excellent environmental adaptability. They can clearly capture the contour details (Zhang et al., 2018a, 2022; Zhang and Yang, 2022a) and spatial distribution of seabed targets (S. Zhao et al., 2023), and they are widely used in marine resource exploration, underwater security and search-and-rescue, and seabed topographic mapping. However, due to complex seabed terrain (Durap, 2025; Zhang and Zhuge, 2024), water scattering (Tan et al., 2019; X. Wang et al., 2026; X. Zhang et al., 2024), noise pollution(Y. Yu et al., 2023; Zhang et al., 2023b) and other interferences(X. Wang et al., 2015; Zhang et al., 2017a, 2018, 2021), the actual sonar images usually exhibit problems such as blurred target textures, low grayscale contrast, uneven target sizes (Xu et al., 2025a; Zhang X. et al., 2019a; Zhang and Ying, 2022d), and indistinct features of weak targets. These issues limit the advancement of detection efficiency and accuracy and significantly raise the challenge of underwater target detection.

Traditional sonar image interpretation methods rely on manual feature extraction and shallow models (Sun et al., 2025; Terven et al., 2023; X. Zhao et al., 2025). In complex and variable marine environments, their representation capability and generalization performance gradually fail to satisfy the demands of high-precision detection (Xu Z. et al., 2025a; Yang, 2024; Xu et al., 2025b). With the breakthrough development of deep learning (Ji and Li, 2026; Liu et al., 2026b; Y. Liu et al., 2026; Vijayakumar and Vairavasundaram, 2024; Zhang et al., 2026a; J. Zheng et al., 2023) and wide application of convolutional neural networks (Di et al., 2026; Qiu and Lin, 2026; T. Wang et al., 2026), a new technical approach has been provided for intelligent interpretation of sonar images. As a typical one-stage framework, the YOLO series achieve excellent performance in image detection fields due to its end-to-end pipeline and high real-time efficiency (Hussain, 2024; Liu et al., 2018; Sapkota et al., 2025; Zhang et al., 2014a). With the advantage of these strengths, YOLO based detection methods have been widely utilized in sidescan sonar image interpretation. Researchers have continuously modified and optimized different YOLO variants to tackle the unique challenges in underwater environments. YOLOv4 (Bochkovskiy et al., 2020; W. Wang and Wang, 2021; X. Zhang et al., n.d.; Zhang et al., 2023a) represents an important milestone. On the basis of YOLOv3 (L. Zhao and Li, 2020), it systematically integrates a variety of optimization approaches for detection, and It achieves an excellent balance between inference efficiency and detectability. Ding et al. developed a lightweight real-time object detection method named YOLOv4-lite for complex scenes(P. Ding et al., 2025). The method replaces the CSPDarknet53 backbone with MobileNetV3 and it incorporates depthwise over-parameterized convolutional layers (DO-Conv) to replace traditional convolutions. This approach significantly reduces the model parameter count to only 62.4% of the original YOLOv4. Additionally, it improves feature extraction performance and strikes a satisfactory balance between accuracy and detecting efficiency. This model sacrifices some detection accuracy. In return, it greatly reduces model size and significantly improves inference speed. It achieves a good balance between speed and accuracy. However, its generalization ability in complex scenes with dense small targets still needs further verification. An enhanced YOLOv4-based method (Fan et al., 2022) for sonar target recognition and categorization was presented by Fan et al. The technique reduces model parameters and network depth by improving the feature extraction network CSPDarknet-53. It substitutes an adaptive spatial feature fusion module (ASFF) for the PANet feature improvement module in order to improve feature fusion. To enhance the receptive field and detection accuracy, the method also adds more fusion feature layers. In terms of detecting quality and efficiency, it performs better than YOLOv4 and YOLOv4-tiny. This model proposes an improved solution for sonar target detection and classification. Its advantage lies in enhancing detection efficiency and accuracy through lightweight design and feature fusion optimization. Its disadvantage is that it still faces difficulties in detecting small targets and identifying irregularly sized targets, and its generalization ability requires further improvement. Yu et al. proposed a shipwreck detection method in SSS images combining YOLOv4 with transfer learning (Yu Y. et al., 2021). The approach firstly implements SSS shipwreck image sample amplification based on SSS imaging mechanisms, and it addresses the challenge of insufficient samples for model construction. Transfer learning is then utilized to introduce weights and universal shipwreck features learned from public datasets into the YOLOv4 network. The experimental results show the constructed shipwreck detection model achieves 85.5% mAP and its detection accuracy is 7.7% higher than that of traditional methods. It demonstrates high-precision shipwreck detection with limited samples. This model effectively solves the problem of insufficient sidescan sonar image samples. However, its data augmentation method remains relatively basic and cannot fully account for all complex imaging factors in actual measurements. Future work still requires more advanced augmentation methods to further improve generalization ability. As the open access YOLOv6 (Geetha, 2024; C. Li et al., 2022; N. Li et al., 2024; Yung et al., 2022), the core features include the design of the EfficientRep backbone and Rep-PAN structure based on the hardware-aware paradigm. By adopting re-parameterization technology with multi-branch training and single-path deployment, it strikes a great compromise between inference efficiency and detection precision. Kalidoss et al. proposed Yolo-STNet. It is an adaptive YOLOv6 model integrated with spatial transformer networks (STNet) (Rajendran et al., 2025). The method addresses problems caused by severe background distortion and unusual object angles in complex environments. In addition to introducing a window function-driven dynamic matching and smooth label technique, it makes use of YOLOv6 advantages. This model adaptively integrates spatial Transformer networks with YOLOv6. It significantly improves detection accuracy and classification capability for multi-angle objects in remote sensing images. It effectively solves the feature distortion problem of traditional models under viewpoint changes. However, its complex structure and high computational cost make deployment on resource-constrained edge devices difficult. It still faces certain challenges in detecting dense small targets. Qu et al. developed VE-YOLOv6. This method is a lightweight small-object detection method designed to tackle the

difficulties of extracting semantic information from small objects while preserving detection speed (J. Wei et al., 2024). It aims to address the difficulty of extracting semantic information from small objects. Meanwhile, it can maintain the detection speed. This method enhances YOLOv6 by designing a lightweight VGNetG based backbone, and it integrates an ECA attention mechanism with expanded position information to preserve detection accuracy. The experimental results based on the public DOTA dataset show that VE-YOLOv6 reduces model computation by 18.91 GFLOPs and parameters by 2.8M, and the mAP only shows a negligible decrease. This model specifically optimizes for small target detection tasks. It effectively reduces model parameters and computational complexity with high detection accuracy for small targets. However, as a conference paper, its robustness and generalization ability in large-scale complex scenarios have not been fully verified. The accuracy loss caused by lightweight design may become more pronounced under extreme conditions. Based on YOLOv6, Wang et al. offered a feature improvement and attention detection head approach to increase underwater object detection (X. Li et al., 2025). To improve multi-scale feature fusion, the technique adds a partial semantic encoding module (PSEM) to the neck network. Additionally, it creates a detecting head with a split dimension weighting head (SDWH). It applies sequential weighting to spatial and channel semantic information. This helps achieve more accurate foreground localization. The outcomes based on the UTDAC2020 and RUOD underwater datasets show that the qualities were improved. The YOLOv6n improved with PSEM and SDWH achieves mAP gains of 3.0% and 3.7%, respectively. This method significantly enhances the detection accuracy of underwater targets affected by insufficient illumination and scattering. Meanwhile, it maintains real-time performance suitable for practical underwater applications. This model significantly improves underwater target detection accuracy. It effectively solves detection challenges in complex underwater environments with low contrast and color distortion. However, its relatively complex structure may affect inference speed. Its generalization ability under extreme underwater lighting conditions still requires further verification. The key features of YOLOv7 (Cao et al., 2023; Wang C.-Y. et al., 2023; Zhao H. et al., 2023; Zhang et al., 2019) include the introduction of an efficient aggregation network, reparameterization techniques, and a dynamic label assignment strategy. It significantly improves the detection accuracy without increasing inference cost. Ge et al. used multibeam forward-looking sonar datasets to create a sophisticated deep learning framework for underwater item detection, and it is adapted from the YOLOv7 architecture (Ge et al., 2025). The framework is uniquely optimized with three factors including the data preprocessing, feature fusion, and loss functions. Compared with existing sonar-based methods, this method exhibits superior target classification performance. This model effectively improves detection robustness and accuracy in complex underwater environments. However, its high computational complexity poses certain challenges for real-time deployment. Its generalization ability across diverse underwater scenarios still requires further validation. For high-precision object detection in sidescan sonar system, Wen et al. suggested an enhanced YOLOv7 model (Wen et al., 2024). This method tackles underwater challenges like sparse target highlights and blurred feature details. Firstly, The Swin-Transformer is presented to improve the model focus on target regions through global modeling and dynamic attention. To further enhance feature representation and accuracy, the convolutional block attention module (CBAM) is employed. Lastly, the YOLOv7 model creatively incorporates a feature scaling factor to handle geometric feature uncertainty. The experiments based on the sidescan sonar datasets show that the improved model achieves 87.9% mAP0.5 and 49.23% mAP0.5:0.95. It represents improvements of 9.28% and 8.41% over the baseline YOLOv7. This model significantly improves detection accuracy for underwater targets in side-scan sonar images. It effectively enhances feature representation of weak targets under complex backgrounds. However, the increased parameter count leads to slower inference speed. Its generalization adaptability across different seabed sediment environments still requires further verification. The multi-gradient feature fusion YOLOv7 model (MFF-YOLOv7) (K. Zheng et al., 2024) was created by Zheng et al. to improve the target recognition capabilities of underwater sonar images. This model integrates the SPPCSPC into YOLOv7 with the multi-scale information fusion module (MIFM). Besides, it uses RFAConv to substitute the standard convolution in the CBS module after ELAN. Furthermore, it incorporates the SCSA mechanism into three intersections where the detecting head and backbone network are connected. The experiments show that MFF-YOLOv7 significantly outperforms other approaches. It exhibits superior underwater object detection performance and stronger generalization potential. Moreover, it offers a more dependable method for identifying targets in challenging underwater settings with low-quality sonar data and few samples. This model effectively improves detection performance for multi-scale targets in underwater sonar images. It particularly alleviates the miss-detection problem of small targets. However, its relatively complex structure leads to high computational costs. Its stability under strong noise interference still requires further verification. YOLOv9 (Chen et al., 2025; X. Liu et al., 2026; C.-Y. Wang et al., 2025) introduces programmable gradient information (PGI) to address the information bottleneck problem in deep networks. Combined with the generalized efficient layer aggregation network (GELAN), it achieves higher detection accuracy while significantly reducing parameters and computational cost. Cui et al. developed an enhanced YOLOv9 model (Cui et al., 2025). The technique enhances detection effectiveness by utilizing an attention mechanism and a small object-focused convolution module. The results based on the public Cylinder2 dataset demonstrate that this method achieves a precision of 71.8% and a recall of 72.7%, with the mAP@50 of 74.3%. It outperforms the original YOLOv9 by 6.5% and YOLO11 by 2.3%, respectively. This model significantly improves small target detection capability in side-scan sonar images. It effectively mitigates the negative impact of degraded underwater images on detection accuracy. However, the image restoration module increases computational overhead. Its robustness under extreme acoustic noise environments still requires further verification. Yuan et al. proposed a novel YOLOv9 Sidescan Network model (Yuan et al., 2025). To improve the model capacity for discriminative learning and operational efficiency, they employ negative sample refinement, an attention-convolution mixed strategy, and spatial and channel reconstruction convolutional layers. The efficiency of feature fusion and object localization is greatly enhanced by the addition of the

bidirectional feature pyramid network and multi-path distance intersection over union metric. Based on results, it achieves 72.1% with mAP@0.5:0.95, and it shows 3.5% improvement over the YOLOv9 baseline model. This model effectively enhances detection accuracy for underwater side-scan sonar targets. It particularly improves the recognition capability of weak targets under complex backgrounds. However, these improvements may increase structural complexity and parameter count, potentially reducing inference efficiency. Its adaptability across different sonar imaging conditions still requires thorough validation. An enhanced YOLOv9-RDA technique (Ma et al., 2026) for forward-looking sonar was presented by Ma et al. It addresses challenges of image degradation and real-time processing demands. The method introduces a lightweight RepViT architecture into the backbone. It successfully lowers the number of parameters and the depth of the model. Without compromising accuracy, it greatly increases inference speed. Furthermore, a DySAU module is intended to improve background modeling and detail reconstruction, while the LSK attention mechanism is employed to maximize small-scale target identification accuracy. Based on results, the mean average precision is improved by 3.2% while the parameter load is reduced by 23%. This model effectively improves detection accuracy and robustness for targets in sonar images. It shows good adaptability particularly in complex marine environments. However, the enhanced modules increase computational complexity, which may affect real-time deployment efficiency. Its generalization ability across different sonar frequency bands still requires further validation.

In summary, this paper comprehensively compares the performances of four object detection algorithms including YOLOv4, YOLOv6, YOLOv7 and YOLOv9. From the theoretical perspective, the network architecture, core innovations and evolutionary context of each approach are systematically compared, and their inherent interrelationships and respective merits are clarified. Based on the publicly available side-scan sonar dataset, the detection ability of the four techniques is compared in this work. Section 2 elaborates on the fundamental principles and key techniques of four YOLO models. Section 3 compares and analyzes their inherent connections and differences in terms of network structure, feature fusion strategies and training schemes. Section 4 introduces the performance evaluation metrics utilized in the experiments. Section 5 provides detailed experimental configurations and comparative result analysis based on the sidescan sonar dataset. The entire work is concluded in section 6, which also explores opportunities for further research.

2 Principle of YOLO

2.1 YOLOv4

YOLOv4 is a milestone model for detection. Its core design philosophy is to elevate detection accuracy with real-time detection speed. It systematically validates and integrates various most effective deep learning techniques available at that time. It attains the ideal ratio of precision to speed. YOLOv4 uses CSPNet (C.-Y. Wang et al., 2020; Wei X. et al., 2024; Q. Yu et al., 2024) strategy to construct CSPDarknet53 (Gouider and Seddik, 2022; Mahasin, n.d.; Senussi and Kang, 2024; S. Zhao et al., 2024) backbone network, and it adds the SPP (Deng et al., 2020; Huang et al., 2026; Kumar et al., 2023; Zhang et al., 2014a; 2014b) and PAN modules in the neck. Meanwhile, it introduces the loss function CIoU_Loss (Du et al., 2021; Gevorgyan, 2022; Huang et al., 2024; X. Liu et al., 2023; X. Wang and Song, 2021).

CSPDarknet53 uses the CSPNet strategy on the basis of Darknet53 by dividing the feature map into two components. The dense convolutional block is used to process one part, while the remaining one is concatenated immediately. This reduces computational complexity and mitigates gradient repetition. It also enhances feature reuse. Meanwhile, It makes it possible for the backbone network to more effectively extract image features. Table 1 lists the framework of CSPDarknet53.

To increase the receptive field and retrieve important contextual information, the SPP module employs multi-scale pooling. It assists in identifying objects that differ greatly in size. YOLOv4 uses PANet (Su et al., 2025; K. Wang et al., 2019; D. Xu et al., 2025) instead of FPN for parameter aggregation. It is adapted for object detection at different levels. The original PANet employs the addition method for feature fusion, and the YOLOv4 algorithm changes the fusion method from addition to concatenation. It is shown in Figure 1.

The CIoU_Loss is a bounding box regression loss function adopted by YOLOv4. It comprehensively considers three core geometric factors, i.e. the overlapping area, the distance between center points, and the aspect ratio, that affect detection accuracy. This makes the regression of predicted boxes more stable and accurate. It accelerates the convergence of the model. It is shown in Equation 1.

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{1}$$

where ρ denotes the Euclidean distance. b and b^{gt} represent the center points of the predicted box and ground-truth box, respectively. c refers to the diagonal length of the minimum circumscribed rectangle of the two boxes. v measures the similarity of aspect ratios. α stands for the weight coefficient. IoU is the fundamental evaluation metric in object detection, it quantifies the overlap degree between predicted boxes and ground-truth boxes. In the loss function of YOLOv4, the weight coefficient α is not a manually tuned hyperparameter. Instead, it serves as a dynamic weight calculated automatically by specific formulas. Its value varies with matching conditions between predicted boxes and ground-truth boxes, including IoU values and aspect ratio differences. The calculation formula of α is presented in Equation 2.

$$\alpha = \frac{v}{(1 - IoU) + v} \tag{2}$$

The calculation formula of v is presented in Equation 3.

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2 \tag{3}$$

Where w^{gt} and h^{gt} denote the width and height of the predicted box. w and h represent the width and height of the ground-truth box. CIoU loss adds an aspect ratio penalty term. It effectively

TABLE 1 CSPDarknet53.

Type	Filters	Size/stride	Output size
Image			416×416×3
Convolutional	32	3×3	416×416×32
Convolutional	64	3×3/2	208×208×64
CSP Block			208×208
Convolutional	32	1×1	
Convolutional	64	3×3	
Residual			208×208×64
Convolutional	128	3×3/2	104×104×128
CSP Block			104×104
Convolutional	64	1×1	
Convolutional	64	3×3	
Residual			104×104×128
Convolutional	128	3×3/2	52×52×256
CSP Block			52×52
Convolutional	128	1×1	
Convolutional	128	3×3	
Residual			52×52×256
Convolutional	512	3×3/2	26×26×512
CSP Block			26×26
Convolutional	256	1×1	
Convolutional	256	3×3	
Residual			26×26×512
Convolutional	1024	3×3/2	13×13×1024
CSP Block			13×13
Convolutional	512	1×1	
Convolutional	512	3×3	
Residual			208×208×64
Global Avgpool		Global	1×1×1024
Connected			1000
Softmax			

alleviates the gradient vanishing problem of IoU loss under low bounding box overlap, and improves fitting accuracy for target shapes. Binary Cross-Entropy (BCE) loss is adopted for both confidence loss and classification loss. Confidence loss distinguishes foreground from background, while classification loss completes multi-label classification tasks.

2.2 YOLOv6

YOLOv6 is an object detection model developed by meituan visual intelligence department. It is designed for industrial applications as a hardware-friendly detector with priority on industrial throughput. Its innovations mainly focus on maximizing performance on industrial-grade GPUs. YOLOv6 improves the core structure with hardware-friendly EfficientRep (Weng et al., 2023; Zhang et al., 2017b; Zhang et al., 2019; Y. Zhang et al., 2025; Zhang et al., 2014b) backbone and RepPAN (Do et al., 2025; Nguyen et al., 2023; G. Wang et al., 2023) neck through structural reparameterization. It adopts anchor-assisted training and self-distillation strategies, and it uses an efficient decoupled head. The developments deliver an excellent balance between high inference throughput and detection accuracy on industrial-grade GPUs.

EfficientRep is designed with hardware awareness. With RepVGG-style (Ding X. et al., 2021; Xu Z. et al., 2025b; Zhang et al., 2026a) reparameterization, it uses multi-branch structures in training to strengthen feature expression. During inference, these branches are converted into single 3×3 convolutions, and it fully utilizes GPU parallel computing. In addition, it uses different designs for models of different sizes. The small models directly stack RepVGGBlock for maximum speed, and the large models use the CSP structure to build more powerful feature extractors. The structure of EfficientRep is shown in Table 2.

The Rep-PAN neck used in YOLOv6 is an enhanced feature fusion module restructured with reparameterization based on PANet. It inherits PANet's bidirectional fusion architecture, and it transmits semantic information top-down and localization information bottom-up. It fuses multi-scale feature maps from different levels of the backbone via upsampling and channel concatenation. The main innovation is that the standard convolutions in PANet are replaced by RepBlocks. As a reparameterized block based on RepVGG, to improve feature representation, RepBlocks employ a multi-branch structure during training. During inference, they are converted into a single 3×3 convolution through structural reparameterization, and it significantly improves computing efficiency on parallel hardware. The Rep-PAN is shown in Figure 2.

YOLOv6 uses SIoU and GIoU (Qian et al., 2023; Yao and Yaoyao, 2020; Z. Zheng et al., 2020) for bounding box regression to regulate network training. YOLOv6-Nano and YOLOv6-Tiny use SIoU. By introducing an angle penalty term, it effectively reduces regression freedom and accelerates model convergence. Meanwhile, the AP can be improved by approximately 0.3%. The small, medium and large models based on YOLOv6 mainly adopt GIoU for regression loss. For large models, GIoU serves as a stable and strong baseline. This can coordinate better with other strategies to maintain overall performance. The SIoU and GIoU are shown in Equations 4, 5, respectively.

$$L_{SIoU} = 1 - IoU + \frac{\Delta + \Omega}{2} \tag{4}$$

$$L_{GIoU} = 1 - IoU + \frac{|C/(B, B^{gt})|}{|C|} \tag{5}$$

In Equation 4, Δ and Ω are the distance and shape losses, respectively. In Equation 5, B and B^{gt} represent the predicted and ground-truth boxes, respectively. C is the minimum enclosing rectangle that contains both $B$and B^{gt}.

2.3 YOLOv7

YOLOv7 uses training strategies including dynamic label assignment (Ge et al., 2024; Xu et al., 2026; Zhang et al., 2022; Zhang, 2024), reparameterized structures (Hoyer et al., 2019; Zhang et al., 2024), and auxiliary head supervision (Yan, 2023; Zhou and Xu, 2024) without

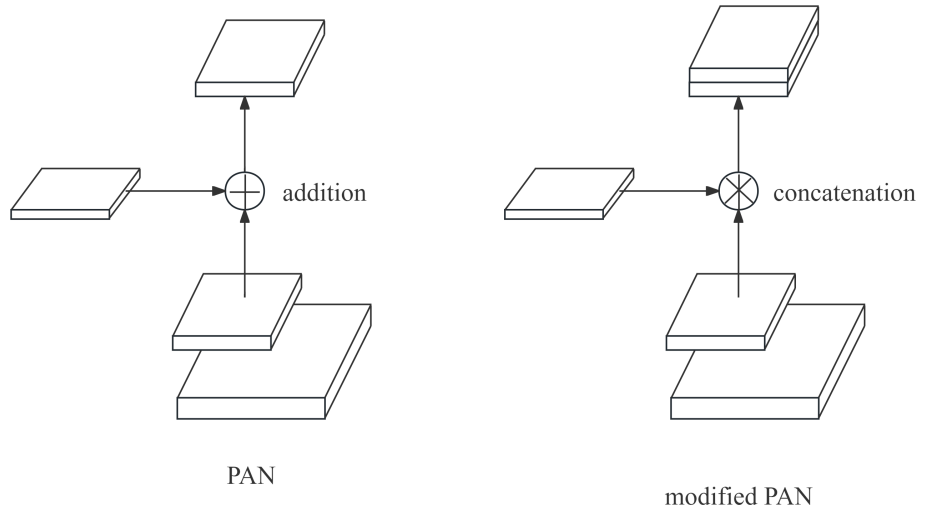


FIGURE 1
Modified PAN.

introducing extra inference cost. It achieves a new balance between accuracy and speed. Using an architectural level, YOLOv7 suggests the extended efficient layer aggregation network (E-ELAN) (Lee et al., 2024; F. Yu et al., 2018). It boosts the feature reuse by controlling gradient paths, and it uses the reparameterized structure (RepConv) (Feng et al., 2024; Liu and Guo, 2025). This method achieves an efficient design with multi-branch training and single-path inference.

Through feature extension and recombination, E-ELAN is intended to improve the network learning capacity without erasing the original gradient routes. The channels and cardinality of computing blocks are increased by a factor of g using group convolutions. It introduces a channel shuffle mechanism to facilitate cross-group information interaction. It finally fuses the expanded multi-group features through element-wise addition. This guides the network to learn richer and more diverse feature representations. This expand–shuffle–merge mechanism maintains the same gradient propagation paths as the original ELAN, and it can effectively improve feature reuse efficiency. As a result, the network obtains stronger feature representation capability without incurring additional inference cost. The E-ELAN is indicated in Figure 3.

TABLE 2 EfficientRep.

Type	Filters	Size/stride	Output size
Image			640×640×3
RepVGGBlock	32	3×3/2	320×320×32
Stage 1			160×160
RepVGGBlock	64	3×3/2	160×160×64
RepVGGBlock	64	3×3	160×160×64
Stage 2			80×80
RepVGGBlock	128	3×3/2	80×80×128
RepVGGBlock	128	3×3	80×80×128
RepVGGBlock	128	3×3	80×80×128
SPPF	128	pool, 5×5	80×80×128
Stage 3			40×40
RepVGGBlock	256	3×3/2	40×40×256
RepVGGBlock	256	3×3	40×40×256
RepVGGBlock	256	3×3	40×40×256
SPPF	256	pool, 5×5	40×40×256
Stage 4			20×20
RepVGGBlock	512	3×3/2	20×20×512
RepVGGBlock	512	3×3	20×20×512
SPPF	512	pool, 5×5	20×20×512

RepConv is featured by convolutional module design that balances accuracy and speed through training-inference decoupling. In order to improve feature representation, it uses a multi-branch structure during training. Through structural reparameterization at inference, the multi-branch structure is equivalently fused into a single-path 3x3 convolution. It can retain the accuracy advantages from training without increasing inference cost. The RepConv is shown in Figure 4.

YOLOv7 makes no fundamental modification to the basic loss formulation. It optimizes the loss optimization path indirectly via an auxiliary training mechanism. It proposes a core Auxiliary Head Training strategy. Multiple auxiliary detection heads are introduced during training to calculate loss together with the primary detection head. This mechanism delivers richer gradient signals to different network layers. The total loss function of YOLOv7 is defined in Equation 6.

$$L_{total} = L_{main} + \lambda \sum_{i=1}^{N} L_{aux}^{(i)} \tag{6}$$

where $\mathcal{L}_{total}$ represents the total loss function. $\mathcal{L}_{main}$ refers to the loss of the main detection head. $\mathcal{L}_{aux}^{(i)}$ stands for the loss of the i-th auxiliary detection head. λ is the weight coefficient for auxiliary loss. N denotes the total number of auxiliary heads. The value of λ is essentially an engineering strategy. It follows a core principle: the weight of auxiliary heads cannot be too large. Each detection head follows a loss composition similar to YOLOv4. It consists of CIoU

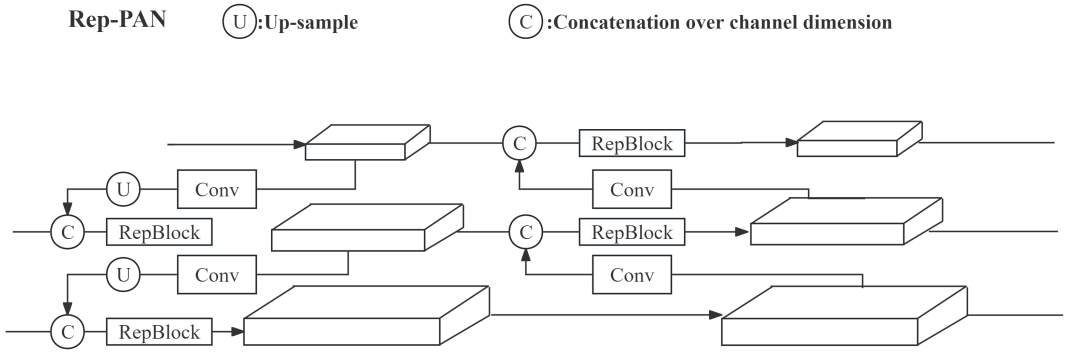


FIGURE 2
Rep-PAN.

regression loss, BCE confidence loss and BCE classification loss. YOLOv7 also innovates the label assignment strategy. It proposes the Coarse-to-Fine Lead Guided Label Assignment method. This method assigns labels progressively from auxiliary heads to the main head. Detection heads at different layers receive supervision signals matching their own feature resolutions. This design keeps the basic loss function unchanged. It optimizes gradient propagation and label matching. It effectively mitigates gradient vanishing and feature mismatch in deep networks. The model achieves faster convergence and higher final accuracy.

2.4 YOLOv9

YOLOv9 is an object detection model released in early 2024. Its core improvements lie in the proposal of programmable gradient information (PGI) (Y. Li et al., 2025; Y. Liu et al., 2025; Mehta et al., 2024; Zhang et al., 2019a) and generalized efficient layer aggregation network (GELAN) (Chandra et al., 2024; Li et al., 2023; Youwai et al., 2024). It aims to address the information bottleneck and gradient vanishing problems that easily occur in deep neural networks during forward propagation. PGI generates reliable gradients by introducing an auxiliary reversible branch. It ensures that key features are not lost during layer-by-layer transmission. This can significantly enhance the model's learning capability without increasing inference cost. GELAN is a lightweight backbone designed for feature fusion, and it improves multi-scale feature fusion efficiency through better parameter utilization and optimized gradient flow. These two innovations improve YOLOv9's detection accuracy without sacrificing real-time speed. The model performs excellently on small objects and in complex environments.

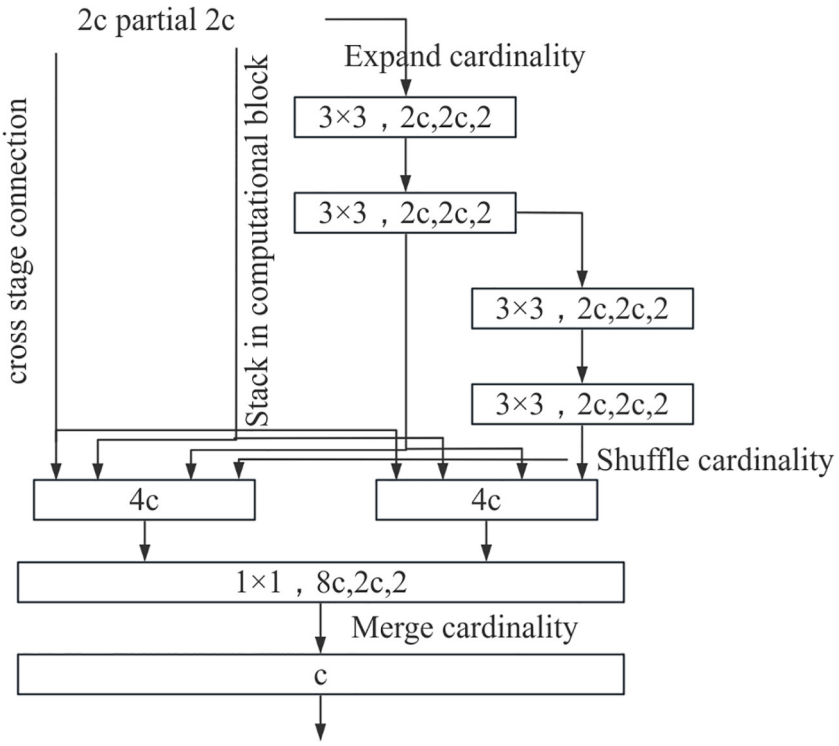


FIGURE 3
E-ELAN.

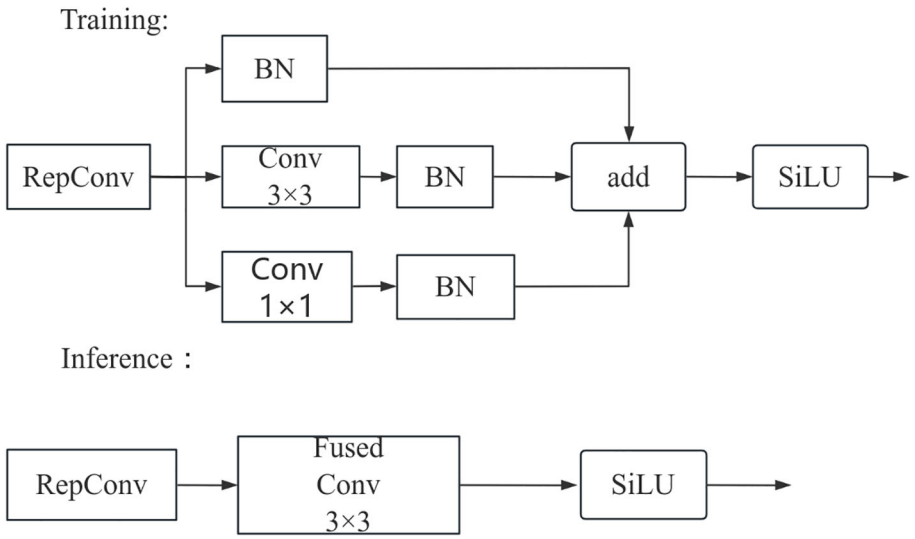


FIGURE 4
RepConv.

The primary branch, the auxiliary reversible branch (Huang and Yang, 2022; Luo et al., 2024; Yang and Liu, 2022; Yaseen, 2024; Zhang, Tang, Zhong, et al., 2014), and multi-level auxiliary information (Huang and Yang, 2022; Liu et al., 2026a; Luo et al., 2024; Yang and Liu, 2022; Yaseen, 2024; Zhang, Tang, Zhong, et al., 2014) are the three basic parts of PGI. The primary one serves as the inference architecture. In order to overcome the information bottleneck caused by deepening networks, the second one uses a reversible topology to produce dependable gradient information. The last one is used to aggregate gradient information from different feature pyramid levels. The fact that PGI only uses the main branch for inference is one of its fundamental advantages. Therefore, it does not introduce any additional inference cost. The PGI is shown in Figure 5.

GELAN combines the advantages of CSPNet and ELAN with generalized improvements. It enables flexible configuration of computational block types, depth, and width according to different computational resource requirements. By optimizing the feature reuse mechanism and gradient propagation paths, GELAN effectively alleviates information loss in deep networks while maintaining extremely low parameters and computational overhead. Its core design philosophy is to enable the network to fully utilize features at each layer without increasing inference burden. This can achieve superior detection performance in lightweight models. The GELAN is shown in Figure 6.

YOLOv9 integrates and improves previous loss function designs. Its two core innovations are Generalized Efficient Layer Aggregation Network (GELAN) and Programmable Gradient Information (PGI). The PGI mechanism builds auxiliary reversible branches and cross-layer gradient propagation paths. It guarantees sufficient and valid gradient signals flow to shallow layers when calculating loss at deep network layers. It addresses gradient vanishing and feature degradation in traditional deep networks. The loss function formula based on PGI is given in Equation 7.

$$L_{total} = L_{main}(y_{main}, y_{gt}) + L_{aux}(y_{aux}, y_{gt}) \quad (7)$$

where y_{main} is the prediction output vector of the main detection head. y_{gt} denotes the ground-truth label. $\mathcal{L}_{aux}$ refers to the loss of the auxiliary detection head. y_{aux} is the prediction output vector of the auxiliary detection head.

3 Comparative analysis of four methods

After elaborating on the core principles of YOLOv4, YOLOv6, YOLOv7, and YOLOv9 in previous sections, this section provides a horizontal comparison of these four algorithms from five aspects including the network architecture, core innovations, training strategies, speed–accuracy trade-offs and application scenarios.

With regard to the network architecture, the traditional CSPDarknet53 serves as the backbone network for YOLOv4, and it introduces the CSP structure. It aims to reduce the computational cost while preserving the integrity of gradient information. YOLOv6 redesigns the architecture and proposes the EfficientRep Backbone and Rep-PAN Neck. It emphasizes the hardware-friendly deployment efficiency. YOLOv7 inherits the CSP idea from YOLOv4 and presents the E-ELAN structure. It enables the deeper networks to learn and converge more effectively by controlling gradient paths. YOLOv9 introduces a novel PGI concept and integrates it with the GELAN architecture. Its main goal is to solve the information bottleneck issue.

The core innovations of these four algorithms reflect an evolutionary trend from modular combination to architecture redesign and then to gradient flow optimization. As a comprehensive integrator, YOLOv4 systematically combines state-of-the-art Bag-of-Freebies and Bag-of-Specials. It validates the effectiveness of

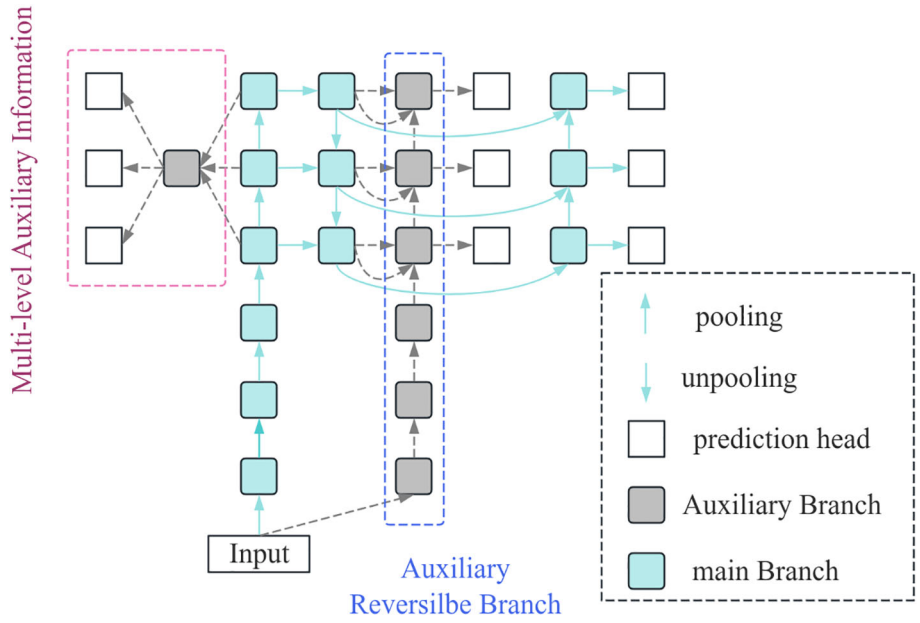


FIGURE 5
PGI.

CSPNet, PANet and other modules. YOLOv6 follows an innovation route oriented toward industrial deployment. It mainly adopts RepVGG-style reparameterization. The network retains a complex multi-branch structure in training, and it is simplified into a single-path structure during inference. Thus, it greatly improves inference speed without accuracy loss. YOLOv7 further optimizes its structure with a dynamic label assignment strategy. This strategy resolves the ambiguity in positive and negative sample allocation during training. Meanwhile, it combines reparameterization and auxiliary head collaborative training. This design improves the detection accuracy without introducing extra inference cost. YOLOv9 focuses on the fundamental problem of information loss and innovatively proposes PGI. The theoretical analysis shows that traditional deep supervision mechanisms have bottlenecks in information transmission. PGI retains complete feed-forward information through auxiliary reversible branches, and it allows lightweight models to achieve high detection accuracy.

In terms of training strategies, YOLOv4 mainly relies on Mosaic data augmentation and the CIoU loss function. YOLOv6 adopts the more advanced Task-Aligned Assigner label assignment strategy and introduces self-distillation to improve the performance of small models. YOLOv7 proposes a mechanism combining coarse and fine label assignment. It solves the issue of scarce positive samples in deep networks by using an auxiliary head to direct the learning process. YOLOv9 inherits the label assignment strategy of YOLOv7. However, its unique PGI mechanism enables auxiliary loss to impact the gradient.

In terms of bounding box regression, YOLOv4 first introduced CIoU loss. On the basis of IoU, it added penalties for center point distance and aspect ratio consistency, solving the gradient vanishing issue of conventional IoU loss when bounding boxes do not overlap at all. YOLOv6 adopts SIoU and GIoU selectively according to model size. Small models use SIoU with an angle penalty for faster convergence, while large models retain the computationally simpler GIoU to maintain training stability. YOLOv7 and YOLOv9 keep the basic form of CIoU for regression loss, yet shift their core innovations to training supervision architectures. YOLOv7 incorporates auxiliary detection heads to deliver extra gradient feedback to shallow features. The weight of auxiliary loss is generally limited to roughly 0.25 times that of the main head loss. Building on YOLOv7, YOLOv9 further proposes the Programmable Gradient Information (PGI) mechanism. Auxiliary reversible branches are constructed to realize programmable gradient propagation throughout the network. Its total loss consists of both main loss and auxiliary loss. The auxiliary branches only participate in loss calculation during training and can be completely removed for inference.

The speed and accuracy form a core trade-off for detection. YOLOv4 established a high-precision benchmark at its time. However, its large parameter size poses challenges for deployment on mobile devices. YOLOv6 shows better performance than YOLOv5 and YOLOv7 under equivalent speed conditions, especially on edge devices. It presents extremely high inference efficiency. YOLOv7 maintains similar inference speed compared to YOLOv6 while delivering higher detection accuracy. In particular, the YOLOv7-Tiny variant exhibits outstanding performance on embedded devices. YOLOv9 realizes the high parameter efficiency with the GELAN architecture. It achieves equivalent accuracy to larger models using fewer parameters, and it creates a new trade-off between lightweight structure and high detection precision.

Based on the above characteristics, the four algorithms focus on different application scenarios. YOLOv4 is suitable for server-side applications with high precision requirements and sufficient

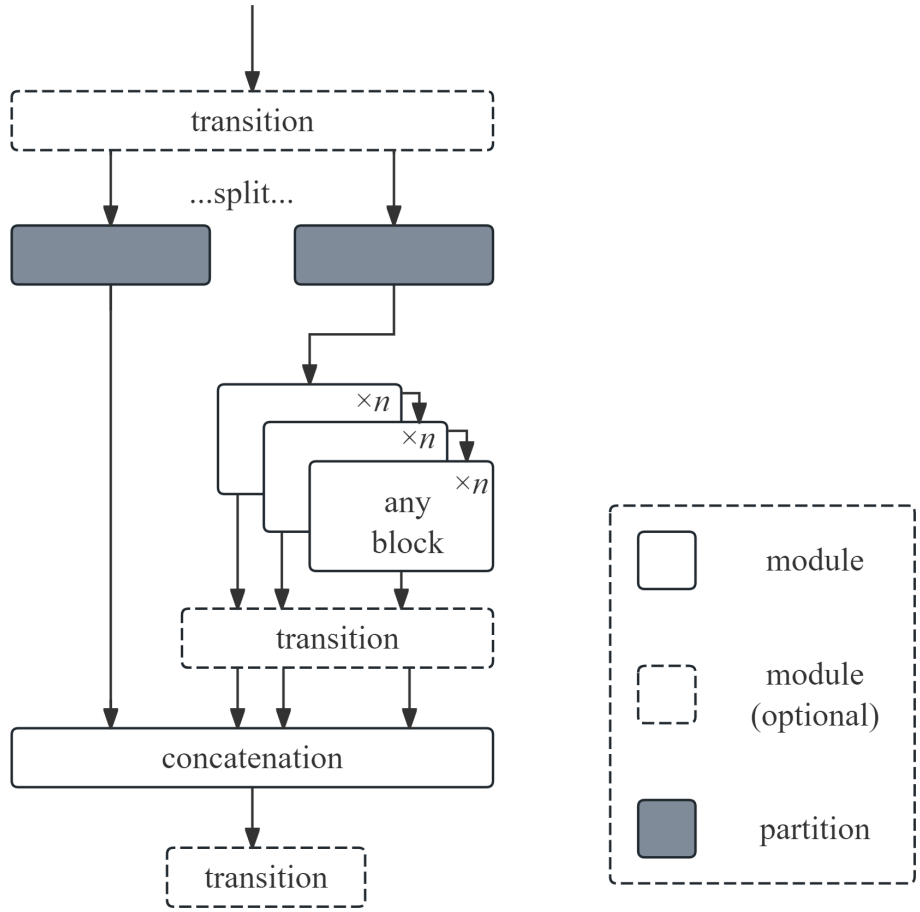


FIGURE 6
GELAN.

computing power. YOLOv6 is ideal for high-throughput scenarios such as industrial inspection and mobile applications due to its excellent hardware adaptability. YOLOv7 is popularly applied in complex scenarios like security monitoring and autonomous driving owing to its flexible model scaling and strong small-object detection capability. YOLOv9 shows great potential in applications with extremely limited resources such as drone imagery or large-model tasks requiring ultra-high precision. The core comparison of YOLOv4, YOLOv6, YOLOv7, and YOLOv9 is shown in Table 3.

4 Performance assessment

4.1 Accuracy assessment

4.1.1 Intersection over union

Intersection over union (IoU) (Nowozin, 2014; Rahman and Wang, 2016; Rezatofighi et al., 2019) is a key indicator, and this parameter is used to measure the overlap of predicted bounding and

TABLE 3 Comparisons of YOLOv4, YOLOv6, YOLOv7, and YOLOv9.

Aspect	YOLOv4	YOLOv6	YOLOv7	YOLOv9
Network Architecture	CSPDarknet53 +SPP + PANet	EfficientRep Backbone+Rep-PAN Neck	E-ELA +RepConv	GELAN + PGI
Core Innovation	Bag-of-Freebies+Bag-of-Specials	Hardware-friendly design +Reparameterization	Dynamic label assignment +Auxiliary head training	Programmable Gradient Information
Training Strategy	Mosaic augmentation+CIoU loss	Task-Aligned Assigner +Selfdistillation	Coarse/fine label assignment +Auxiliary head guidance	PGI retains complete gradient flow+inherits YOLOv7 label assignment
Speed and Accuracy	High accuracy but large parameters;high deployment cost	Excellent inference efficiency; industrial deployment friendly	Good balance; Tiny version excels on edge devices	Extremely high parameter efficiency; minimal accuracy loss under lightweight settings
Application Scenario	Server-side high precision detection	Industrial inspection, mobile apps, high-throughput scenarios	Security surveillance, autonomous driving, complex scenes	Drone view, Resource constrained devices, lightweight high-accuracy needs

ground-truth bounding boxes. It is the ratio of the intersection and union areas, and it is within the range [0, 1]. The formula for IoU is given in Equation 8.

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{8}$$

where A and B are predicted bounding and ground-truth bounding boxes, individually.

4.1.2 Precision and recall

The precision (Juba and Le, 2019; Kynkäänniemi et al., 2019; Zhang et al., 2024; Zhang et al., 2022b) represents the ratio of true positive samples to all samples predicted as positive. It shows how effectively the model can prevent false targets. The recall represents the ratio of correctly detected positive samples to all actual positive samples, and it demonstrates the model capacity to prevent missed detections. The formulas for Precision P and Recall R are given in Equations 9, 10, respectively.

$$P = \frac{TP}{TP + FP} \tag{9}$$

$$R = \frac{TP}{TP + FN} \tag{10}$$

where TP (true positive) refers to the number of positive samples that were successfully detected. FP (false positive) is the number of mistakenly negative samples, and FN (false negative) is the number of omitted positive samples.

4.1.3 Average precision

The average precision (AP) refers to the area obtained by integrating or interpolating the precision-recall (P-R) curve. It comprehensively evaluates the model performance by combining precision and recall. The formula for AP is given in Equation 11.

$$\text{AP} = \int_0^1 P(R)dR \tag{11}$$

4.1.4 Mean average precision

The fundamental metric for assessing the overall performance of object detection models is the mean average precision (mAP), which is the average precision across all categories. For example, N categories are exited. The i category precision is AP_i, and then mAP is given by Equation 12.

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N}\text{AP}_i \tag{12}$$

4.2 Efficiency evaluation metrics

4.2.1 Inference speed

FPS (Frames Per Second) serves as the core metric to evaluate the inference speed of YOLO models. It represents the number of images the model can process per second. In object detection, FPS directly reflects the real-time performance of a model. A higher FPS means the model can process more image frames within a fixed time unit. Such models are more suitable for deployment in delay-sensitive application scenarios. The formula of FPS is presented in Equation 13. *Latency* is the end-to-end time for a single image from input to output.

$$\text{FPS} = \frac{1}{Latency} \tag{13}$$

4.2.2 Complexity

The computational cost (FLOPs) and the number of parameters (Params) are the primary metrics used to assess the complexity of a model. The number of parameters expressed in millions (M), indicates the total number of trainable parameters in the model file. It directly affects the memory usage and storage size occupied by the model. Floating point operations (FLOPs), which are measured in billions (G), are typically referred to as floating-point operations. The FLOPs reflect the computational resources required for model inference and serve as an important reference for evaluating the theoretical speed of the model. We suppose that the size of the input feature map is $H_{in} \times W_{out}$. The input and output channel numbers are C_{in} and C_{out}, respectively. The convolution kernel size is $K \times K$. Then the formulas for the number of parameters and FLOPs of this convolutional layer are given in Equations 14, 15, respectively.

$$\text{Params} = K \times K \times C_{in} \times C_{out} \tag{14}$$

$$\text{FLOPs} = K \times K \times C_{in} \times C_{out} \times H_{out} \times W_{out} \tag{15}$$

Considering the output feature map, the height is denoted by H_{out}, and W_{out} denotes the width. By comparing these metrics, the ability of the model to balance accuracy and efficiency can be comprehensively evaluated.

5 Experiments and results

5.1 Experimental setup

Two datasets are adopted in the experiments. Dataset D1 contains 664 images annotated with three target categories: person, shipwreck, and plane. Without any data augmentation expansion, it serves as a small-sample original dataset for verifying model overfitting. Dataset D2 is constructed on the basis of Dataset D1. Data expansion is realized by collecting new side-scan sonar images and applying geometric transformations. The dataset finally contains 2838 samples with target categories identical to those of D1, it is adopted to compare the alleviation effect of model overfitting relative to the small-sample Dataset D1. To ensure the engineering reference value of the comparison results, all models were tested on the same RTX 4090 hardware platform with identical dataset splits. Considering the inherent differences among YOLOv4, YOLOv6, YOLOv7, YOLOv9, YOLOv13n and YOLO26n in data

augmentation strategies, anchor box mechanisms and optimizer scheduling schemes, all models were trained strictly with the default hyperparameters provided by their respective official repositories. No model-specific hyperparameter tuning was conducted to fairly evaluate the generalization performance of each algorithm under standard default configurations.

5.2 Experimental results

This study uses six object detection models, including YOLOv4, YOLOv6, YOLOv7, YOLOv9, YOLOv13n and YOLO26n. It carries out comparative experiments on underwater object detection for sidescan sonar images. For visual comparison and verification, 16 sample images are chosen from each of the two datasets. This research adopts seven core evaluation metrics. They are Precision (P), Recall (R), mAP@0.5, mAP50–95, frames per second (FPS), floating point operations (FLOPs) and model parameter quantity. This study combines qualitative visual detection results and quantitative numerical data. It fully compares and analyzes the comprehensive performance of different models in the underwater object detection task of sidescan sonar images.

Figures 7–18 show visual detection results clearly. All models achieve higher target confidence and more complete bounding boxes on Dataset D2 than Dataset D1. YOLOv4 is a traditional heavy model with many parameters. It performs poorly on noisy sonar images from D1. It misses many faint human targets and small shipwrecks. Striped seabed clutter and reef textures are often misidentified as underwater targets, and they produce many useless bounding boxes. On clean D2 samples, target confidence rises greatly. Missing and false detections drop sharply, yet redundant boxes still exist. Lightweight YOLOv13n and YOLO26n have stronger robustness for low-quality sonar images. Even on D1 samples with low contrast and heavy background noise, they output complete, high-confidence target boxes steadily. They have far fewer duplicate boxes, empty detections and background misjudgments than large models like YOLOv4 and YOLOv7. Quantitative tables quantify the performance gaps seen in visual results. As is shown in Tables 4, 5. mAP@0.5 measures overall detection ability under an IoU threshold of 0.5. With this loose threshold, YOLOv6 on D1 reaches the highest value of 0.944. mAP50–95 covers multiple IoU thresholds from 0.5 to 0.95. It fairly reflects the model's ability to regress fine outlines and locate targets precisely. This metric directly affects practical tasks such as mapping shipwreck outlines and estimating target sizes. YOLOv9 on D2 achieves the best mAP50–95 of 0.573. For lightweight deployment and real-time inference, YOLO26n has obvious overall advantages. It only has 2.375M parameters and a FLOPs cost of 5.2G. It reaches an inference speed of 526.32 FPS when tested on D2, the fastest among all compared models. YOLOv4 is a bulky model with 64M parameters and 127.2G FLOPs. Its heavy computation limits its speed to only 12.94 FPS. It cannot meet the real-time processing requirements of underwater equipment and has an obvious weakness in inference speed.

By comparing each model across D1 and D2, the two datasets have very different data distributions. The gaps between their metrics clearly show the generalization ability of each network. It can be analyzed overfitting during training based on how much performance shifts between the two datasets. Overfitting means the model memorizes unique textures, noise and target shapes from the training set. It achieves high accuracy on samples similar to training data. But its performance drops sharply on images with different distributions and unknown noise patterns. A larger metric gap between the two datasets indicates stronger overfitting. Deep heavy models like YOLOv4 and YOLOv7 show severe overfitting. For YOLOv4, mAP@0.5 rises from 0.664 on D1 to 0.887 on D2. Its mAP50–95 increases by nearly 0.15. Visual results confirm that many faint targets with different shapes and noise from training samples are totally undetectable on D1. Large backbones contain massive parameters. They easily memorize fixed seabed clutter and gray-scale patterns in training images. They fail to extract general underwater target features and have poor generalization. YOLOv6 is also highly sensitive to dataset distribution shifts. It shows an opposite trend: its accuracy on D1 is higher than on D2. Its Recall falls from 0.869 on D1 to 0.732 on D2. This model overfits clear target features in D1. It learns many image-specific details instead of universal target outlines. It generates far more missed detections

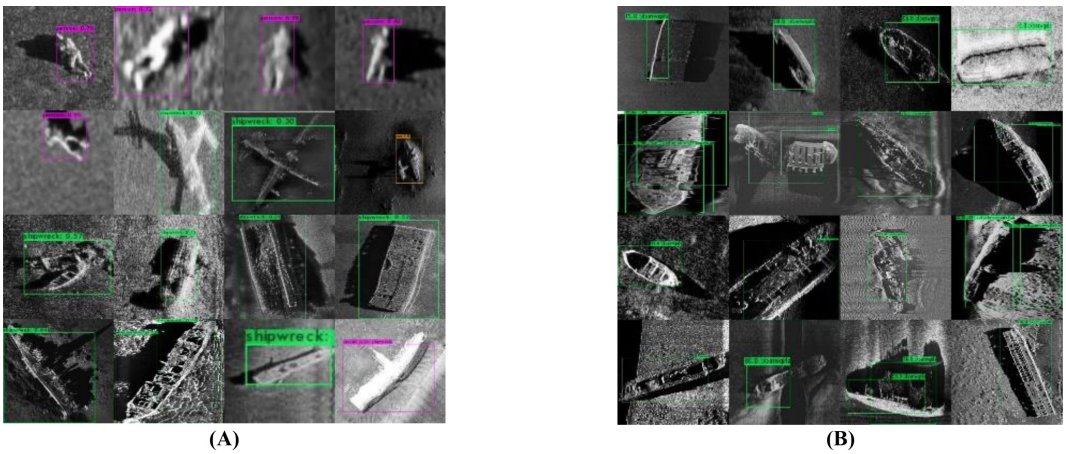


FIGURE 7
YOLOv4-D1. **(A)** Visualization of YOLOv4-D1 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv4-D1 detection results (16 samples, Group 2).

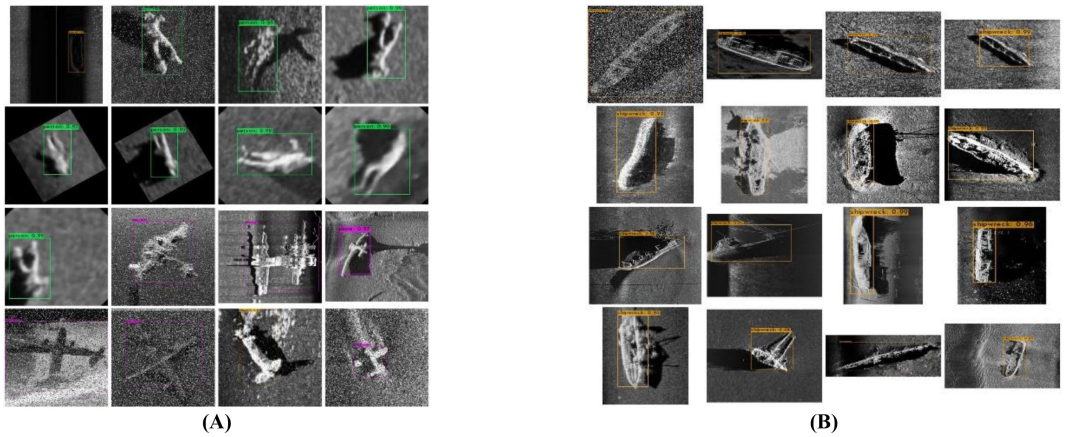


FIGURE 8
YOLOv4-D2. **(A)** Visualization of YOLOv4-D2 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv4-D2 detection results (16 samples, Group 2).



FIGURE 9
YOLOv6-D1. **(A)** Visualization of YOLOv6-D1 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv4-D1 detection results (16 samples, Group 2).

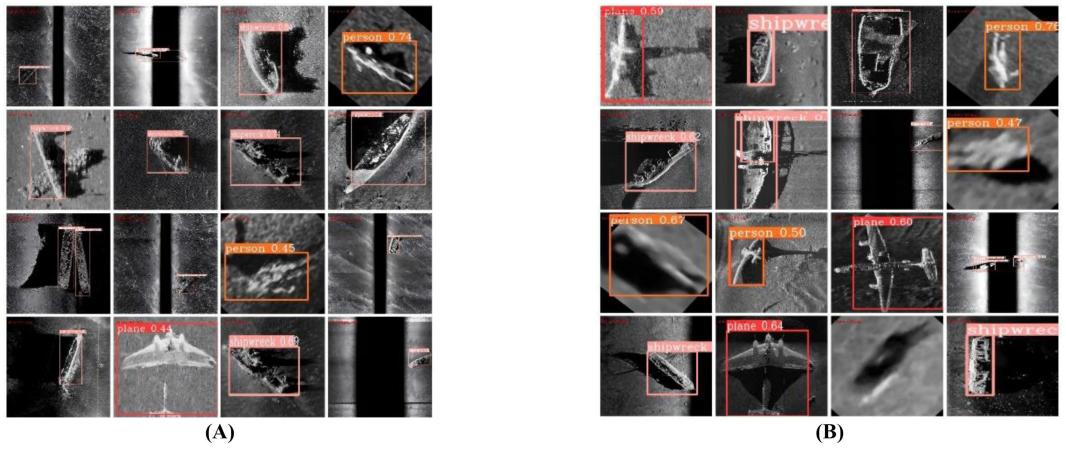


FIGURE 10
YOLOv6-D2. **(A)** Visualization of YOLOv6-D2 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv6-D2 detection results (16 samples, Group 2).

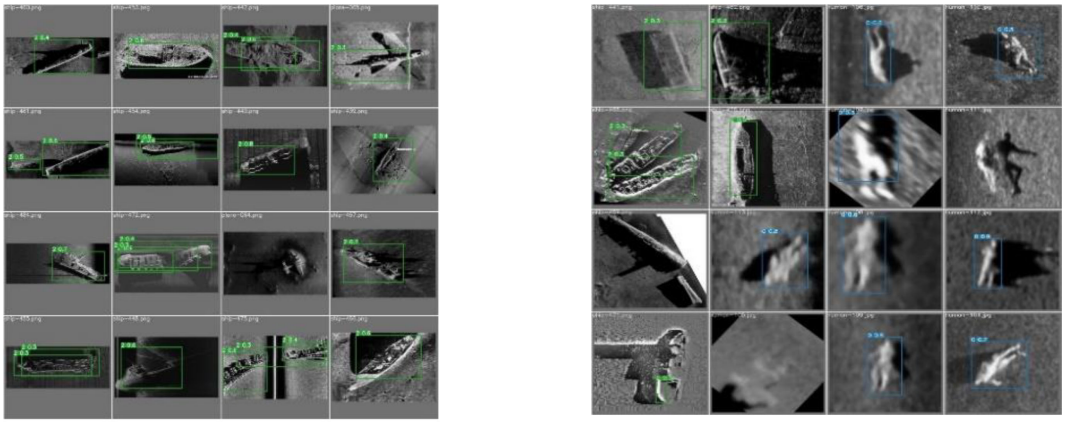


FIGURE 11
YOLOv7-D1. **(A)** Visualization of YOLOv7-D1 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv7-D1 detection results (16 samples, Group 2).

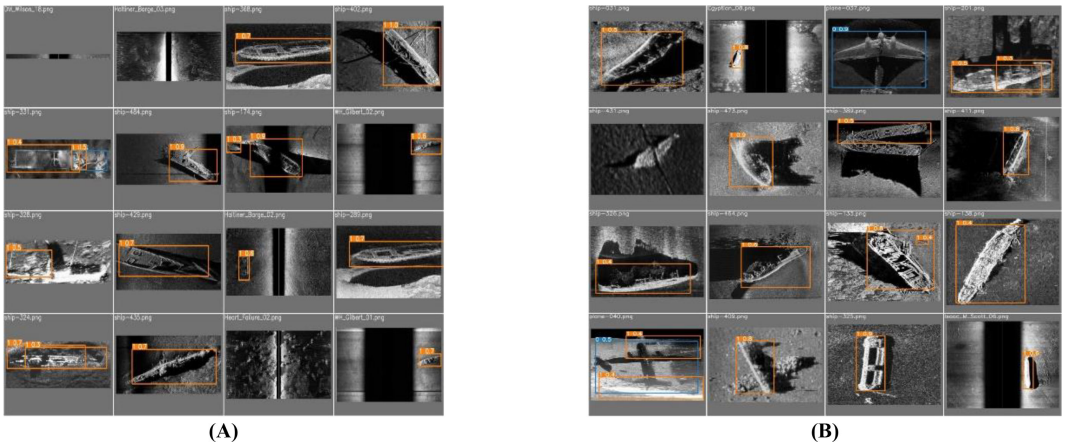


FIGURE 12
YOLOv7-D2. **(A)** Visualization of YOLOv7-D2 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv7-D2 detection results (16 samples, Group 2).

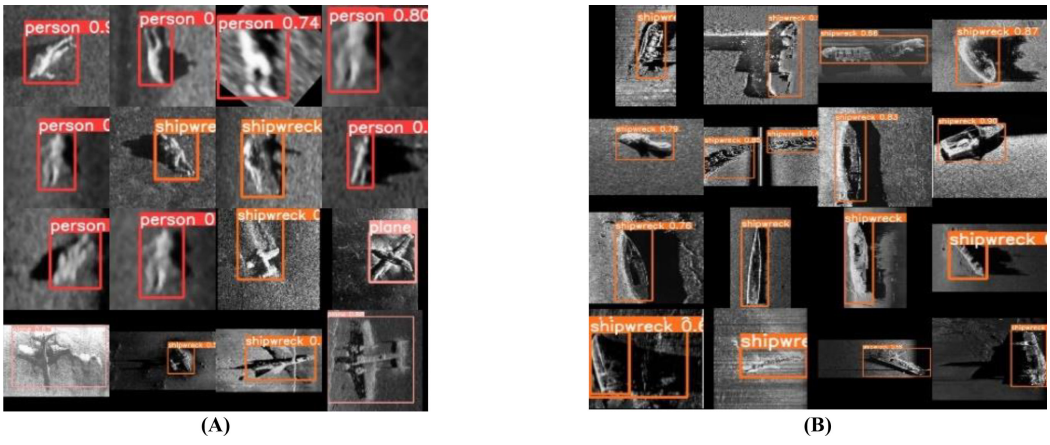


FIGURE 13
YOLOv9-D1. **(A)** Visualization of YOLOv9-D1 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv9-D1 detection results (16 samples, Group 2).

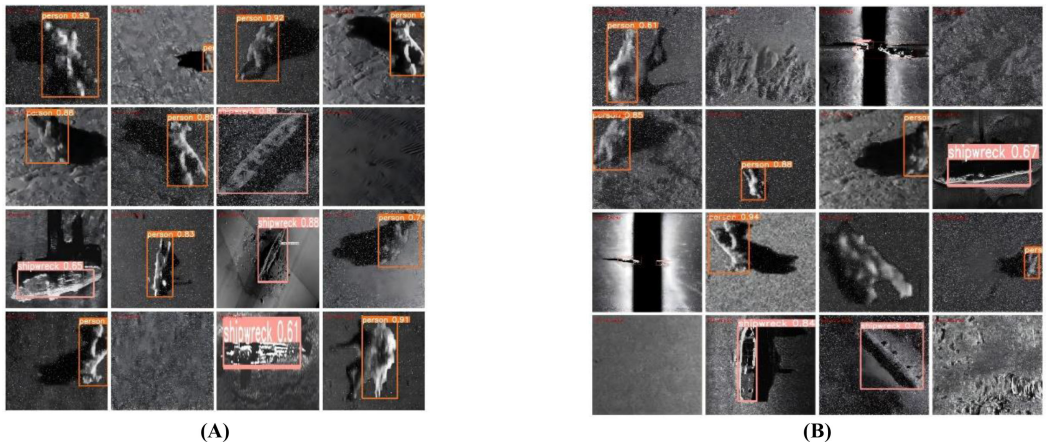


FIGURE 14
YOLOv9-D2. **(A)** Visualization of YOLOv9-D2 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv9-D2 detection results (16 samples, Group 2).

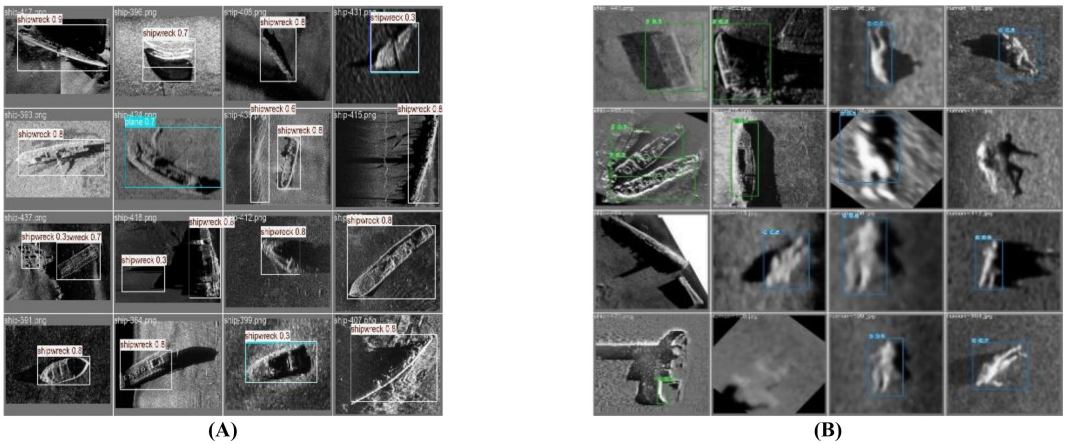


FIGURE 15
YOLOv13n-D1. **(A)** Visualization of YOLOv13n-D1 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv13n-D1 detection results (16 samples, Group 2).

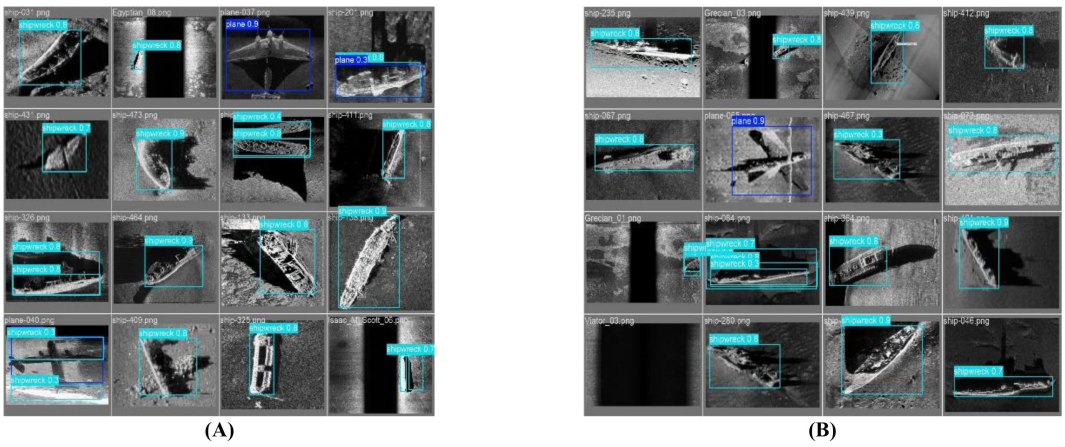


FIGURE 16
YOLOv13n-D2. **(A)** Visualization of YOLOv13n-D2 detection results (16 samples, Group 1). **(B)** Visualization of YOLOv13n-D2 detection results (16 samples, Group 2).

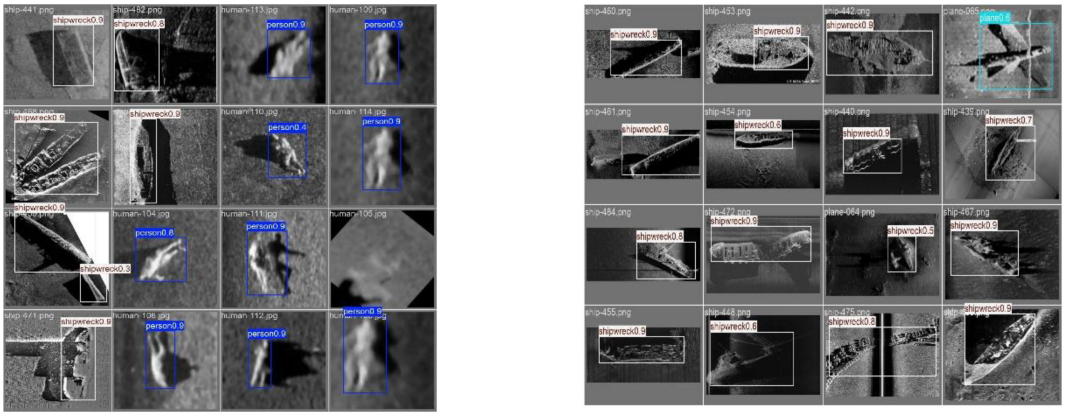


FIGURE 17
YOLO26n-D1. **(A)** Visualization of YOLO26n-D1 detection results (16 samples, Group 1). **(B)** Visualization of YOLO26n-D1 detection results (16 samples, Group 2).

when transferred to images with different imaging styles. Lightweight models YOLOv9, YOLOv13n and YOLO26n behave differently. All their Precision, Recall and mAP gaps between D1 and D2 stay below 0.08. Target confidence changes mildly in visual results. None of them fail massively on either dataset. Lightweight networks have fewer channels and parameters. Their limited parameter space cannot fully store redundant background noise and irrelevant textures from training data. Their structure naturally restrains overfitting. Among them, ultra-light YOLO26n and YOLOv13n simplify feature extraction modules further. During training, they focus more on general geometric shapes and gray-scale rules of shipwrecks and human targets, rather than unique local noise. Their performance drops the least when imaging styles and noise levels change. They achieve the best generalization and anti-overfitting performance.

This paper evaluates all models from five aspects: visual detection performance, accuracy metrics (Precision and Recall), real-time inference FPS, model parameters and FLOPs, as well as anti-overfitting generalization ability. The evaluation clearly defines suitable underwater sidescan sonar application scenarios and limits for each model. It provides references for model selection in practical projects with different hardware and detection tasks. YOLOv4 is only suitable for offline laboratory analysis with sufficient computing power. It can reach high Recall when hardware resources are abundant, image preprocessing is complete, and all samples have consistent imaging quality. However, it has huge model size and computational cost. It cannot be deployed on embedded devices such as shipborne portable equipment and small underwater vehicles. Besides, it suffers severe overfitting. It often misses a large number of targets on sonar images from wild and unfamiliar sea areas, so it cannot support real-time marine detection. YOLOv6 fits semi-offline batch processing tasks with unified image collection standards and basic target detection demands. Its inference speed is moderate. But it poorly adapts to noisy, low-contrast degraded sonar images and lacks generalization. It cannot be used for real-time underwater search and rescue or large-area sea survey. YOLOv7 serves as a compromise choice for platforms with medium computing power. It is lighter than

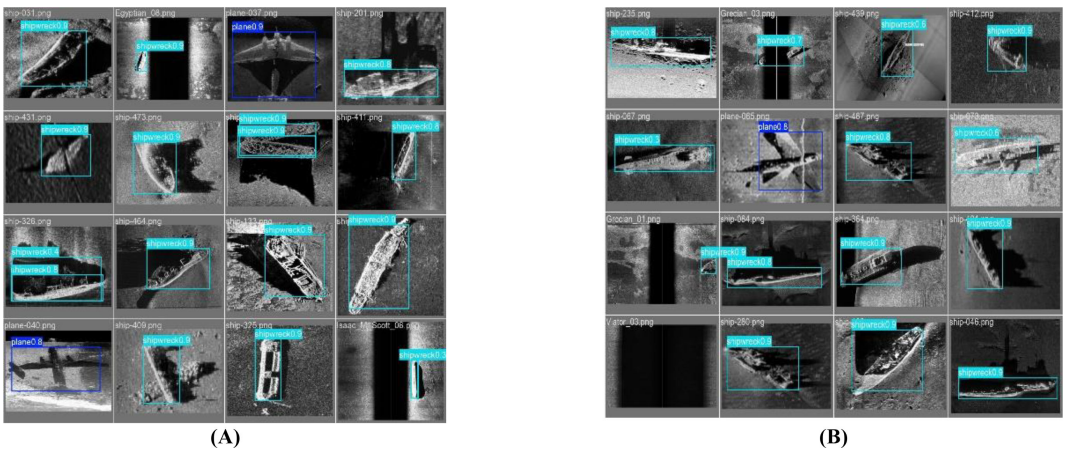


FIGURE 18
YOLO26n-D2. **(A)** Visualization of YOLO26n-D2 detection results (16 samples, Group 1). **(B)** Visualization of YOLO26n-D2 detection results (16 samples, Group 2).

TABLE 4 Quantitative detection metrics of various YOLO models on side-scan sonar D1 and D2 datasets.

Model - dataset	Class	P	R	mAP@0.5	mAP50~95	FPS	FLOPs (G)	Params (M)
YOLOv4-D1	all	0.708	0.692	0.664	0.387	12.94	127.2	64
YOLOv4-D2	all	0.912	**0.936**	0.887	0.538	27.43	127.2	64
YOLOv6-D1	all	**0.927**	0.869	**0.944**	0.425	184.86	45.17	18.5
YOLOv6-D2	all	0.891	0.732	0.854	0.484	349.39	45.17	18.5
YOLOv7-D1	all	0.774	0.733	0.734	0.353	112.79	51.32	37.6
YOLOv7-D2	all	0.820	0.815	0.860	0.501	222.19	51.32	37.6
YOLOv9-D1	all	0.839	0.792	0.862	0.430	139.97	48.65	32.1
YOLOv9-D2	all	0.885	0.863	0.915	**0.573**	285.53	48.65	32.1
YOLOv13n-D1	all	0.877	0.921	0.925	0.562	188.68	6.2	2.448
YOLOv13n-D2	all	0.821	0.836	0.870	0.548	416.67	6.2	2.448
YOLO26n-D1	all	0.874	0.841	0.892	0.550	204.08	**5.2**	**2.375**
YOLO26n-D2	all	0.855	0.824	0.885	0.556	**526.32**	**5.2**	**2.375**

Bold values means values with the best performance.

YOLOv4 and balances basic detection accuracy and inference speed. Yet it still has a large number of parameters and weak generalization under complex seabed clutter. It is only recommended for regular coastal underwater surveys with fixed target sizes and stable sea conditions. YOLOv9 has unique advantages in localization accuracy under multiple IoU thresholds. It achieves the highest mAP50–95 among all models and regresses target outlines precisely. It fits specialized tasks requiring high bounding box accuracy, such as underwater archaeology, precise size measurement of shipwreck debris and target contour reconstruction. It can run on mid-to-high-end industrial controllers, but its computation cost is higher than ultra-light models. It cannot be deployed on miniature AUVs with limited computing power for real-time detection. YOLOv13n is a balanced lightweight model. It obtains excellent Recall and effectively reduces missed targets. It has low hardware deployment requirements and matches medium and small autonomous underwater vehicles for real-time marine search and rescue and rapid large-area sea surveys. It strikes a good balance between detection accuracy and inference speed. YOLO26n delivers the best overall engineering performance in this comparative experiment. It has the smallest parameter size and FLOPs, and far faster inference speed than other networks. Meanwhile, it keeps high Precision, Recall and mAP values. It performs well against clutter interference and overfitting. It adapts to all kinds of low-computing embedded terminals, including portable handheld side-scan sonars, miniature underwater vehicles and small buoy detection systems. It can run continuous long-term real-time underwater detection, making it the optimal detection architecture for practical marine engineering applications.

6 Conclusion

This paper focuses on underwater targets in sidescan sonar images. Two datasets are constructed: D1 with heavy noise and complex sea conditions, and D2 with high signal-to-noise ratio and clear images. Six detection models YOLOv4, YOLOv6, YOLOv7, YOLOv9, YOLOv13n and YOLO26n are tested for comparison. This study combines visual detection results and seven quantitative metrics about accuracy, speed and lightweight degree. It systematically analyzes detection performance, generalization, overfitting rules and applicable engineering scenarios, and draws core conclusions for practical engineering reference.

First, the signal-to-noise ratio of sonar images is a key external factor affecting underwater target detection. Model size directly determines its tolerance to complex seabed clutter. The traditional heavy model YOLOv4 costs massive computation and runs very slowly. It only achieves decent detection results on high-quality sonar images. It causes many missed targets and false alarms on low-contrast, noisy samples. Lightweight YOLOv13n and YOLO26n adopt simplified network structures. They stably detect underwater targets under poor imaging conditions and show much stronger robustness against clutter. Different models have different strengths. YOLOv6 achieves the best overall detection performance under loose IoU thresholds. YOLOv9 excels at precise target outline localization. YOLO26n has the smallest parameters and computation cost, and delivers the fastest inference speed. It offers irreplaceable lightweight advantages for real-time embedded deployment.

Second, model parameter size is positively correlated with overfitting risk. Lightweight networks have better ability to adapt to different data distributions by nature. Large-capacity YOLOv4 and YOLOv7 easily memorize unique seabed textures and gray features from training data. Their detection accuracy varies greatly between the two datasets, leading to weak generalization. YOLOv6 overfits high-signal-to-noise samples, and its Recall drops sharply under noisy conditions. Limited by small parameter capacity, YOLOv9, YOLOv13n and YOLO26n cannot store redundant background noise. They focus more on learning general geometric features of shipwrecks and human targets during training. Their metrics fluctuate less on sonar images with different noise levels, with milder overfitting. Side-scan sonar images have single gray levels and repeated seabed noise. Expanding datasets from multiple sea areas and adding online augmentations such as noise overlay

TABLE 5 Metrics of each model for different categories.

Model - dataset	Class	P	R	mAP@0.5	mAP50~95	FPS	FLOPs (G)	Params (M)
YOLOv4-D1	plane	0.571	1.00	0.449	0.263	12.94	127.2	64
	shipwreck	0.705	0.632	0.740	0.434	12.94	127.2	64
	person	0.769	0.869	0.802	0.470	12.94	127.2	64
YOLOv4-D2	plane	0.800	0.762	0.781	0.458	27.43	127.2	64
	shipwreck	0.882	0.865	0.914	0.536	27.43	127.2	64
	person	0.950	0.910	0.966	0.566	27.43	127.2	64
YOLOv6-D1	plane	1.000	0.819	**0.942**	0.301	184.86	45.17	18.5
	shipwreck	0.829	0.880	0.920	0.411	184.86	45.17	18.5
	person	1.000	0.900	0.923	0.546	184.86	45.17	18.5
YOLOv6-D2	plane	0.909	0.476	0.743	0.351	349.39	45.17	18.5
	shipwreck	0.857	0.769	0.846	0.445	349.39	45.17	18.5
	person	0.908	0.952	0.971	0.657	349.39	45.17	18.5
YOLOv7-D1	plane	0.750	0.500	0.534	0.178	112.79	51.32	37.6
	shipwreck	0.573	0.800	0.745	0.334	112.79	51.32	37.6
	person	0.896	0.788	0.965	0.539	112.79	51.32	37.6
YOLOv7-D2	plane	0.809	0.619	0.761	0.386	222.19	51.32	37.6
	shipwreck	0.765	0.859	0.850	0.457	222.19	51.32	37.6
	person	0.885	0.968	0.970	0.661	222.19	51.32	37.6
YOLOv9-D1	plane	0.776	0.667	0.698	0.280	139.97	48.65	32.1
	shipwreck	0.844	0.920	0.923	0.472	139.97	48.65	32.1
	person	0.952	0.909	0.971	0.561	139.97	48.65	32.1
YOLOv9-D2	plane	0.808	0.810	**0.836**	**0.487**	285.53	48.65	32.1
	shipwreck	0.889	0.869	0.928	0.542	285.53	48.65	32.1
	person	0.957	0.910	**0.981**	**0.689**	285.53	48.65	32.1
YOLOv13n - D1	plane	0.731	0.846	0.816	0.468	188.68	6.2	2.448
	shipwreck	0.919	0.918	0.963	0.593	188.68	6.2	2.448
	person	0.981	1.000	**0.995**	**0.624**	188.68	6.2	2.448
YOLOv13n - D2	plane	0.689	0.667	0.713	0.416	416.67	6.2	2.448
	shipwreck	0.851	0.877	**0.929**	**0.558**	416.67	6.2	2.448
	person	0.922	0.963	0.968	0.672	416.67	6.2	2.448
YOLO26n - D1	plane	0.796	0.692	0.763	0.450	204.08	5.2	2.375
	shipwreck	0.919	0.932	**0.966**	**0.600**	204.08	5.2	2.375
	person	0.908	0.900	0.947	0.599	204.08	5.2	2.375
YOLO26n - D2	plane	0.759	0.667	0.767	0.472	526.32	5.2	2.375
	shipwreck	0.871	0.864	0.913	0.545	526.32	5.2	2.375
	person	0.937	0.941	0.974	0.650	526.32	5.2	2.375

Bold values means values with the best performance.

and gray transformation can effectively reduce performance gaps across sea conditions and mitigate accuracy loss in field detection.

Third, this paper divides differentiated application scopes for each YOLO model in underwater sonar detection by comprehensively considering detection accuracy, inference speed, hardware deployment cost and generalization stability. The results provide clear guidelines for practical model selection. YOLOv4 is only fit for offline image analysis on high-computing platforms. YOLOv6 and YOLOv7 are only suitable for batch offline surveys with stable imaging and medium computing power. YOLOv9 serves high-precision specialized tasks requiring accurate outline localization, such as underwater archaeology and wreck mapping. YOLOv13n balances low missed detection rate and lightweight design, matching medium and small underwater vehicles for large-area rapid scanning. YOLO26n has the best overall practical performance. It works on all kinds of low-computing portable sonar devices and miniature underwater vehicles, making it the top choice for long-term real-time marine detection.

In summary, this study verifies that lightweight YOLO networks better meet the practical deployment demands of side-scan sonar underwater target detection. It clarifies the trade-offs among accuracy, real-time performance and generalization between heavy and lightweight models. It provides complete experimental support for model selection of underwater sonar detection systems with different hardware platforms and detection tasks. Future work will expand sonar datasets of underwater targets from multiple sea areas and scales. We will specially optimize lightweight backbone feature extraction networks and embed noise suppression modules, to further improve detection accuracy of faint small targets in deep sea with heavy clutter and boost cross-sea-area generalization performance.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

DL: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. HC: Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. JC: Data curation, Formal Analysis, Writing – original draft, Writing – review & editing. YD: Data curation, Formal Analysis, Writing – original draft, Writing – review & editing. JZ: Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research was supported by the "High-Quality Scientific and Technological Achievement Award Cultivation Project of Yancheng Institute of Technology" (Project No.: CGJ2025022). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript, the Natural Science Foundation of Hunan Province for Excellent Young Scholars (2025JJ40036), and the Changsha Municipal Distinguished Young Scientists and Technologists Program (kq2506001), which are granted by Jiahua Zhu.

Acknowledgments

No permits were required for the work carried out in this study, which complied with all relevant regulations. The authors are grateful to D. Liu, J. Chen, Y. Ding and J. Zhu for their assistance with documentation, mapping, and graphics.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

Bochkovskiy, A., Wang, C.-Y., and Liao, H.-Y. M. (2020). YOLOv4: optimal speed and accuracy of object detection. *arXiv*. 2004.10934. doi: 10.48550/arXiv.2004.10934

Cao, L., Zheng, X., and Fang, L. (2023). The semantic segmentation of standing tree images based on the Yolov7 deep learning algorithm. *Electronics*. 12, 929. doi: 10.3390/electronics12040929

Chandra, N., Vaidya, H., Sawant, S., and Meena, S. R. (2024). A novel attention-based generalized efficient layer aggregation network for landslide detection from satellite data in the higher Himalayas, Nepal. *Remote Sens.* 16, 2598. doi: 10.3390/rs16142598

Chen, C., Lee, H., and Chen, M. (2025). Steel surface defect detection method based on improved YOLOv9. *Sci. Rep.* 15, 25098. doi: 10.1038/s41598-025-10647-1

Chen, W., Lan, F., Cai, B., Wang, M., Chen, D., and Zhao, T. (2026). D2C-SID: A divergence to convergence strategy for sonar image denoising. *IEEE Trans. Multimedia* 28, 6352–6364. doi: 10.1109/TMM.2026.3668561

Chen, X., Yuan, M., Yang, Q., Yao, H., and Wang, H. (2023). Underwater-YCC: underwater target detection optimization algorithm based on YOLOv7. *J. Mar. Sci. Eng.* 11, 995. doi: 10.3390/jmse11050995

Cui, X., Zhang, J., Zhang, L., Zhang, Q., and Han, J. (2025). Small object detection in side-scan sonar images based on SOCA-YOLO and image restoration. *Front. Mar. Sci.* 12. doi: 10.3389/fmars.2025.1542832

Deng, J., Xuan, X., Wang, W., Li, Z., Yao, H., and Wang, Z. (2020). A review of research on object detection based on deep learning. *J. Physics: Conf. Ser.* 1684, 12028. doi: 10.1088/1742-6596/1684/1/012028

Di, J., Wang, Q., and Yang, Y. (2026). Efficient transformer-based UAV image object detection network based on convolutional self-attention and matchability-adaptive classification. *J. King Saud Univ. Comput. Inf. Sci.* 38, 251. doi: 10.1007/s44443-026-00621-w

Ding, P., Li, T., Qian, H., Ma, L., and Chen, Z. (2025). A lightweight real-time object detection method for complex scenes based on YOLOv4. *J. Real-Time Image Process.* 22, 68. doi: 10.1007/s11554-025-01645-7

Ding, X., Zhang, X., Ma, N., Han, J., Ding, G., and Sun, J. (2021).RepVGG: Making VGG-Style ConvNets great again. Available online at: https://openaccess.thecvf.com/content/CVPR2021/html/Ding_RepVGG_Making_VGG-Style_ConvNets_Great_Again_CVPR_2021_paper.html (Accessed May 26, 2026).

Do, J., Kim, S., Youk, G., Lee, J., and Kim, M. (2025).PAN-Crafter: Learning modality-consistent alignment for PAN-sharpening. Available online at: https://openaccess.thecvf.com/content/ICCV2025/html/Do_PAN-Crafter_Learning_Modality-Consistent_Alignment_for_PAN-Sharpening_ICCV_2025_paper.html (Accessed April 15, 2026).

Du, S., Zhang, B., Zhang, P., and Xiang, P. (2021). "An improved bounding box regression loss function based on CIOU loss for multi-scale object detection", in: *2021 IEEE 2nd International Conference on Pattern Recognition and Machine Learning (PRML)* (Chengdu: IEEE), 92–98. doi: 10.1109/PRML52754.2021.9520717

Durap, A. (2025). Explainable machine learning for bathymetric mapping: Adaptive normalization and feature engineering in complex seabed terrains. *Ocean Sci. J.* 60, 52. doi: 10.1007/s12601-025-00251-9

Fan, X., Lu, L., Shi, P., and Zhang, X. (2022). A novel sonar target detection and classification algorithm. *Multimedia Tools Appl.* 81, 10091–10106. doi: 10.1007/s11042-022-12054-4

Feng, S., Qian, H., Wang, H., and Wang, W. (2024). $$η$$-repyolo: Real-time object detection method based on $$η$$-RepConv and YOLOv8. *J. Real-Time Image Process.* 21, 81. doi: 10.1007/s11554-024-01462-4

Ge, L., Singh, P., and Sadhu, A. (2025). Advanced deep learning framework for underwater object detection with multibeam forward-looking sonar. *Struct. Health Monit.* 24, 1991–2007. doi: 10.1177/14759217241235637

Ge, L., Wang, G., Zhang, T., Zhuang, Y., Chen, H., Dong, H., et al. (2024). Adaptive dynamic label assignment for tiny object detection in aerial images. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 17, 6201–6214. doi: 10.1109/JSTARS.2024.3379515

Geetha, A. S. (2024). What is YOLOv6? a deep insight into the object detection model. *arXiv*. 2412.13006. doi: 10.48550/arXiv.2412.13006

Gevorgyan, Z. (2022). SIoU loss: more powerful learning for bounding box regression. *arXiv*. 2205.12740. doi: 10.48550/arXiv.2205.12740

Gouider, C., and Seddik, H. (2022). "YOLOv4 enhancement with efficient channel recalibration approach in CSPdarknet53", in: *2022 IEEE Information Technologies & Smart Industrial Systems (ITSIS)* (Paris: IEEE), 1–6. doi: 10.1109/ITSIS56166.2022.10118431

Hoyer, S., Sohl-Dickstein, J., and Greydanus, S. (2019). Neural reparameterization improves structural optimization. *arXiv*. 1909.04240. doi: 10.48550/arXiv.1909.04240

Huang, P., Fan, Y., Teng, X., Tan, J., and Guo, L. (2026). Range-doppler algorithm of multireceiver synthetic aperture sonar using nonuniform signal. *PloS One* 21, e0343983. doi: 10.1371/journal.pone.0343983

Huang, P., Tian, S., Su, Y., Tan, W., Dong, Y., and Xu, W. (2024). IA-CIOU: An improved IOU bounding box loss function for SAR ship target detection methods. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 17, 10569–10582. doi: 10.1109/JSTARS.2024.3402540

Huang, P., and Yang, P. (2022). Synthetic aperture imagery for high-resolution imaging sonar. *Front. Mar. Sci.* 9. doi: 10.3389/fmars.2022.1049761

Hussain, M. (2024). YOLOv1 to v8: Unveiling each variant–A comprehensive review of YOLO. *IEEE Access* 12, 42816–42833. doi: 10.1109/ACCESS.2024.3378568

Ji, C., and Li, Y. (2026). UWLight-YOLO: An adaptive lightweight attention network for underwater sonar image object detection. *Cognit. Rob.* 6, 78–91. doi: 10.1016/j.cogr.2026.01.001

Juba, B., and Le, H. S. (2019). "Precision-recall versus accuracy and the role of large data sets", in: *Proceedings of the AAAI Conference on Artificial Intelligence* (AAAI Press), 33, 4039–4048. doi: 10.1609/aaai.v33i01.33014039

Khan, A., Fouda, M. M., Do, D.-T., Almaleh, A., Alqahtani, A. M., and Rahman, A. U. (2024). Underwater target detection using deep learning: Methodologies, challenges, applications, and future evolution. *IEEE Access* 12, 12618–12635. doi: 10.1109/ACCESS.2024.3353688

Kumar, A., Kalia, A., Sharma, A., and Kaushal, M. (2023). A hybrid tiny YOLO v4-SPP module based improved face mask detection vision system. *J. Ambient Intell. Hum. Comput.* 14, 6783–6796. doi: 10.1007/s12652-021-03541-x

Kynkäänniemi, T., Karras, T., Laine, S., Lehtinen, J., and Aila, T. (2019). "Improved precision and recall metric for assessing generative models," in *Advances in Neural Information Processing Systems*, vol. 32. (Vancouver: Curran Associates, Inc.). Available online at: https://proceedings.neurips.cc/paper/2019/hash/0234c510bc6d908b28c70ff313743079-Abstract.html (Accessed May 3, 2026).

Lee, J., Choi, Y. H., Lee, H., Park, H. J., Hong, J. S., Lee, J. H., et al. (2024). Enhanced swine behavior detection with YOLOs and a mixed efficient layer aggregation network in real time. *Animals*. 14, 3375. doi: 10.3390/ani14233375

Li, Y., Hong, Y., Wang, Z., Chen, J., Liu, R., Ding, S., et al. (2025b). Nonlinear multi-head cross-attention network and programmable gradient information for gaze estimation. *Sci. Rep.* 15, 27135. doi: 10.1038/s41598-025-12466-w

Li, C., Li, L., Jiang, H., Weng, K., Geng, Y., Li, L., et al. (2022). YOLOv6: a single-stage object detection framework for industrial applications. *arXiv*. 2209.02976. doi: 10.48550/arXiv.2209.02976

Li, N., Wang, M., Yang, G., Li, B., Yuan, B., and Xu, S. (2024). DENS-YOLOv6: A small object detection model for garbage detection on water surface. *Multimedia Tools Appl.* 83, 55751–55771. doi: 10.1007/s11042-023-17679-7

Li, G., Xiong, C., Qian, G., Thabet, A., and Ghanem, B. (2023). DeeperGCN: Training deeper GCNs with generalized aggregation functions. *IEEE Trans. Pattern Anal. Mach. Intell.* 45, 13024–13034. doi: 10.1109/TPAMI.2023.3306930

Li, X., Zhao, Y., Su, H., Wang, Y., and Chen, G. (2025a). Efficient underwater object detection based on feature enhancement and attention detection head. *Sci. Rep.* 15, 5973. doi: 10.1038/s41598-025-89421-2

Lin, Y., Song, L., Feng, Q., He, S., Huang, N., and Zhu, Y. (2026). DFSE-YOLO: Deep feature selective enhancement for efficient shipwreck detection in side-scan sonar imagery. *Sci. Rep.* 16, 3227. doi: 10.1038/s41598-025-33164-7

Liu, W., and Guo, X. (2025). RRFNet: A free-anchor brain tumor detection and classification network based on reparameterization technology. *PloS One* 20, e0325483. doi: 10.1371/journal.pone.0325483

Liu, Y., Ji, Y., Ren, Q., Shi, B., Liu, N., Lu, M., et al. (2025). An innovative small-target detection approach against information attenuation: Fusing enhanced programmable gradient information and a novel Mamba module. *Sensors* 25. doi: 10.3390/s25072117

Liu, D., Jin, Z., Chen, J., and Xu, Z. (2026a). Detection of submarine pipeline and cable targets based on depth feature of high resolution sonar image. *PloS One* 21, e0346343. doi: 10.1371/journal.pone.0346343

Liu, D., Jin, Z., Chen, J., and Xu, Z. (2026b). Seafloor sediment detection with sidescan sonar image based on YOLO11 and YOLO26. *Electron. Lett.* 62, e70650. doi: 10.1049/ell2.70650

Liu, L., Li, H., Zhu, J., Peng, Y., and Zhai, G. (2026c). YOLO variant evaluation and transfer learning analysis for side-scan sonar object detection. *J. Mar. Sci. Eng.* 14, 550. doi: 10.3390/jmse14060550

Liu, K., Sun, Q., Sun, D., Peng, L., Yang, M., and Wang, N. (2023). Underwater target detection based on improved YOLOv7. *J. Mar. Sci. Eng.* 11, 677. doi: 10.3390/jmse11030677

Liu, C., Tao, Y., Liang, J., Li, K., and Chen, Y. (2018). "Object detection based on YOLO network", in: *2018 IEEE 4th Information Technology and Mechatronics Engineering Conference (ITOEC)* (Chongqing: IEEE), 799–803. doi: 10.1109/ITOEC.2018.8740604

Liu, X., Wu, G., Zhou, T., Wang, Z., Wang, K., Li, J., et al. (2026d). SCS-YOLOv9: A novel YOLOv9 framework for small object detection via edge computing. *Int. J. Mach. Learn. Cybern.* 17, 99. doi: 10.1007/s13042-025-02848-x

Liu, Y., Yan, G., Chen, T., and Huo, G. (2026e). SSGA-YOLO: A lightweight sonar image object detection network with efficient convolution and acoustic-aware attention for embedded systems. *IET Image Proc.* 20, e70313. doi: 10.1049/ipr2.70313

Liu, X., Yang, X., Chen, Y., and Zhao, S. (2023). Object detection method based on CIoU improved bounding box loss function. *Chin. J. Liquid Crystals Displays* 38, 656–665. doi: 10.37188/CJLCD.2022-0282

Luo, F., Bian, W., Jie, B., Dong, H., and Fu, X. (2024). ARBFPN-YOLOv8: Auxiliary reversible bidirectional feature pyramid network for UAV small target detection. *Signal. Image Video Process.* 19, 63. doi: 10.1007/s11760-024-03661-9

Ma, X., Yang, M., Zhang, Z., and Zang, C. (2026). Enhanced YOLOv9-RDA for target detection in sonar images. *Ships Offshore Struct.* 1–15. doi: 10.1080/17445302.2026.2627524

Mahasin, M. (2022). Comparison of CSPDarkNet53, CSPResNeXt-50, and EfficientNet-B0 backbones on YOLO V4 as object detector. *Int. J. Eng. Sci. Inf. Technol.*. 2 (3), 64–72. doi: 10.52088/IJESTY.V1I4.291

Mehta, M., Gunavathi, C., Mathews, N., and Ruby, D. (2024). Can programmable gradient information enhance polyp segmentation? *IEEE Access* 12, 169277–169290. doi: 10.1109/ACCESS.2024.3496999

Nguyen, K. H., Nguyen, H. V. N., Tran, H. N., and Quach, L.-D. (2023). "Combining autoencoder and Yolov6 model for classification and disease detection in chickens", in: *Proceedings of the 2023 8th International Conference on Intelligent Information Technology (ICIIT 2023)* (ACM), 132–138. doi: 10.1145/3591569.3591591

Nowozin, S. (2014).Optimal decisions from probabilistic models: The intersection-over-union case. Available online at: https://openaccess.thecvf.com/content_cvpr_2014/html/Nowozin_Optimal_Decisions_from_2014_CVPR_paper.html (Accessed April 22, 2026).

Qian, X., Zhang, N., and Wang, W. (2023). Smooth GIoU loss for oriented object detection in remote sensing images. *Remote Sens.* 15, 1259. doi: 10.3390/rs15051259

Qiao, J., Yu, J., Huang, Y., Feng, H., Jia, D., Wang, Z., et al. (2026). AUVs for seabed surveying: A comprehensive review of side-scan sonar-based target detection. *J. Mar. Sci. Eng.* 14, 145. doi: 10.3390/jmse14020145

Qiu, Y., and Lin, Z. (2026). Lightweight model LMW-YOLO for small object detection in remote sensing images. *Sci. Rep*. 16, 11644. doi: 10.1038/s41598-026-45055-6

Rahman, M. A., and Wang, Y. (2016). "Optimizing intersection-over-union in deep neural networks for image segmentation," in *Advances in Visual Computing*. Eds. G. Bebis, R. Boyle, B. Parvin, D. Koracin, F. Porikli and S. Skaff (Cham: Springer International Publishing), 234–244. doi: 10.1007/978-3-319-50835-1_22

Rajendran, G. B., Srinivasan, G. T., and Rathakrishnan, N. (2025). Adaptive YOLOv6 with spatial transformer networks for accurate object detection and multi-angle classification in remote sensing images. *Expert Syst. Appl.* 282, 127796. doi: 10.1016/j.eswa.2025.127796

Rezatofighi, H., Tsoi, N., Gwak, J., Sadeghian, A., Reid, I., and Savarese, S. (2019).Generalized intersection over union: A metric and a loss for bounding box regression. Available online at: https://openaccess.thecvf.com/content_CVPR_2019/html/Rezatofighi_Generalized_Intersection_Over_Union_A_Metric_and_a_Loss_for_CVPR_2019_paper.html (Accessed May 10, 2026).

Sapkota, R., Flores-Calero, M., Qureshi, R., Badgujar, C., Nepal, U., Poulose, A., et al. (2025). YOLO advances to its genesis: A decadal and comprehensive review of the You Only Look Once (YOLO) series. *Artif. Intell. Rev.* 58, 274. doi: 10.1007/s10462-025-11253-3

Senussi, M. F., and Kang, H.-S. (2024). Occlusion removal in light-field images using CSPDarknet53 and bidirectional feature pyramid network: a multi-scale fusion-based approach. *Appl. Sci.*. 14, 9332. doi: 10.3390/app14209332

Su, J., Xu, D., Qiu, L., Xu, Z., Lin, L., and Zheng, J. (2025). A high-accuracy underwater object detection algorithm for synthetic aperture sonar images. *Remote Sens.*. 17, 2112. doi: 10.3390/rs17132112

Sun, S., Xu, Z., Cao, X., Zheng, J., Yang, J., and Jin, N. (2025). A high-performance and lightweight maritime target detection algorithm. *Remote Sens.*. 17, 1012. doi: 10.3390/rs17061012

Tan, C., Zhang, X., Yang, P., and Sun, M. (2019). A novel sub-bottom profiler and signal processor. *Sensors*. 19, 5052. doi: 10.3390/s19225052

Terven, J., Córdova-Esparza, D.-M., and Romero-González, J.-A. (2023). A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS. *Mach. Learn. Knowl. Extr.* 5, 1680–1716. doi: 10.3390/make5040083

Vijayakumar, A., and Vairavasundaram, S. (2024). YOLO-based object detection models: A review and its applications. *Multimedia Tools Appl.* 83, 83535–83574. doi: 10.1007/s11042-024-18872-y

Wang, C.-Y., Bochkovskiy, A., and Liao, H.-Y. M. (2023).YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. Available online at: https://openaccess.thecvf.com/content/CVPR2023/html/Wang_YOLOv7_Trainable_Bag-of-Freebies_Sets_New_State-of-the-Art_for_Real-Time_Object_Detectors_CVPR_2023_paper.html?utm_source=chatgpt.com (Accessed April 8, 2026).

Wang, C.-Y., Liao, H.-Y. M., Wu, Y.-H., Chen, P.-Y., Hsieh, J.-W., and Yeh, I.-H. (2020).CSPNet: A new backbone that can enhance learning capability of CNN. Available online at: https://openaccess.thecvf.com/content_CVPRW_2020/html/w28/Wang_CSPNet_A_New_Backbone_That_Can_Enhance_Learning_Capability_of_CVPRW_2020_paper.html (Accessed May 18, 2026).

Wang, K., Liew, J. H., Zou, Y., Zhou, D., and Feng, J. (2019).PANet: Few-shot image semantic segmentation with prototype alignment. Available online at: https://openaccess.thecvf.com/content_ICCV_2019/html/Wang_PANet_Few-Shot_Image_Semantic_Segmentation_With_Prototype_Alignment_ICCV_2019_paper.html (Accessed April 29, 2026).

Wang, T., Lu, J., Wan, B., Lu, R., Sun, Y., Wu, D., et al. (2026). Feature boosting and scale-aware network with multi-modal information for underwater salient object detection. *Eng. Appl. Artif. Intell.* 174, 114489. doi: 10.1016/j.engappai.2026.114489

Wang, G., Pan, H., Wang, H., Geng, H., Lan, J., and Han, L. (2023). "PUPose: real-time pose estimation for push-up on device", in: *Proceedings of the 2022 6th International Conference on Computer Science and Artificial Intelligence (CSAI 2022)* (ACM), 21–27. doi: 10.1145/3577530.3577534

Wang, X., and Song, J. (2021). ICIoU: Improved loss based on complete intersection over union for bounding box regression. *IEEE Access* 9, 105686–105695. doi: 10.1109/ACCESS.2021.3100414

Wang, W., and Wang, Y. (2021). "Underwater target detection system based on YOLO v4", in: *2021 2nd International Conference on Artificial Intelligence and Information Systems (ICAIIS 2021)* (ACM), 1–5. doi: 10.1145/3469213.3470310

Wang, X., Wang, Z., and Bi, D. (2026). Dual-modal vision–sonar object detection for underwater robots based on deep learning. *J. Mar. Sci. Eng.*. 14, 338. doi: 10.3390/jmse14040338

Wang, C.-Y., Yeh, I.-H., and Mark Liao, H.-Y. (2025). "YOLOv9: Learning what you want to learn using programmable gradient information," in *Computer Vision – Eccv 2024*. Eds. A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler and G. Varol (Cham: Springer Nature Switzerland), 1–21. doi: 10.1007/978-3-031-72751-1_1

Wang, X., Zhang, X., and Zhu, S. (2015). "Upsampling based back projection imaging algorithm for multi-receiver synthetic aperture sonar", in: *2015 International Industrial Informatics and Computer Engineering Conference (IIICEC 2015)* (Atlantis Press), 1610–1615. doi: 10.2991/iiicec-15.2015.355

Wei, J., Qu, Y., Gong, M., Ma, Y., and Zhang, X. (2024). "VE-YOLOv6: a lightweight small target detection algorithm", in: *2024 4th International Conference on Neural Networks, Information and Communication Engineering (NNICE)* (Guangzhou: IEEE), 873–876. doi: 10.1109/NNICE61279.2024.10498732

Wei, X., Wang, F., He, D., Liu, C., and Xu, D. (2024). Coal gangue image recognition model based on CSPNet-YOLOv7 target detection algorithm. *COAL Sci. Technol.* 52, 238–248. doi: 10.12438/cst.2023-0546

Wen, X., Wang, J., Cheng, C., Zhang, F., and Pan, G. (2024). Underwater side-scan sonar target detection: YOLOv7 model combined with attention mechanism and scaling factor. *Remote Sens.*. 16, 2492. doi: 10.3390/rs16132492

Weng, K., Chu, X., Xu, X., Huang, J., and Wei, X. (2023). EfficientRep: an efficient RepVGG-style ConvNets with hardware-aware neural network design. *arXiv*. 2302.00386. doi: 10.48550/arXiv.2302.00386

Xu, D., He, Y., Su, J., Qiu, L., Lin, L., Zheng, J., et al. (2025). An ultra-lightweight and high-precision underwater object detection algorithm for SAS images. *Remote Sens.*. 17, 3027. doi: 10.3390/rs17173027

Xu, Z., Xu, D., He, Y., Lin, L., and Zheng, J. (2025a). A hybrid filtering method with no-reference quality assessment for synthetic aperture sonar images. *PloS One* 20, e0332458. doi: 10.1371/journal.pone.0332458

Xu, Z., Xu, D., Lin, L., Song, L., Song, D., Sun, Y., et al. (2025b). Integrated object detection and communication for synthetic aperture radar images. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 18, 294–307. doi: 10.1109/JSTARS.2024.3495023

Xu, D., Xu, Z., Lin, L., Zheng, J., Song, L., and Ji, Z. (2026). JMUCOD: Joint maritime-underwater cross-domain object detection algorithm. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 19, 564–579. doi: 10.1109/JSTARS.2025.3635474

Yan, Z. (2023).Auxiliary supervision for regularizing deep learning based image classification. Available online at: https://theses.lib.polyu.edu.hk/handle/200/12643 (Accessed May 21, 2026).

Yang, P. (2024). An imaging algorithm for high-resolution imaging sonar system. *Multimedia Tools Appl.* 83, 31957–31973. doi: 10.1007/s11042-023-16757-0

Yang, P., and Liu, J. (2022). "Effect of non-uniform sampling on sonar focusing", in: *2022 14th International Conference on Communication Software and Networks (ICCSN)* (Chongqing: IEEE), 109–113. doi: 10.1109/ICCSN55126.2022.9817582

Yao, L., and Yaoyao, Q. (2020). "Insulator detection based on GIOU-YOLOv3", in: *2020 Chinese Automation Congress (CAC)* (Shanghai: IEEE), 5066–5071. doi: 10.1109/CAC51589.2020.9326959

Yaseen, M. (2024). What is YOLOv9: An in-depth exploration of the internal features of the next-generation object detector. *arXiv*. 2409.07813. doi: 10.48550/arXiv.2409.07813

Youwai, S., Chaiyaphat, A., and Chaipetch, P. (2024). YOLO9tr: A lightweight model for pavement damage detection utilizing a generalized efficient layer aggregation network and attention mechanism. *J. Real-Time Image Process.* 21, 163. doi: 10.1007/s11554-024-01545-2

Yu, Y., Li, Y., Gong, Q., and Li, Y. (2021). Shipwreck detection in side-scan sonar images using YOLOv4 with transfer learning. *Haiyang Cehui* 41, 38–42.

Yu, F., Wang, D., Shelhamer, E., and Darrell, T. (2018).Deep layer aggregation. Available online at: https://openaccess.thecvf.com/content_cvpr_2018/html/Yu_Deep_Layer_Aggregation_CVPR_2018_paper.html (Accessed April 12, 2026).

Yu, Q., Xie, S., Xu, K., Luo, H., and Zhou, X. (2024). "CSPNet: a lightweight and efficient convolutional neural network for fault diagnosis", in: *Proceedings of the 2024 4th International Conference on Control and Intelligent Robotics (ICCIR 2024)* (ACM), 201–206. doi: 10.1145/3687488.3687524

Yu, Y., Zhao, J., Huang, C., and Zhao, X. (2023). Treat noise as domain shift: Noise feature disentanglement for underwater perception and maritime surveys in side-scan sonar images. *IEEE Trans. Geosci. Remote Sens.* 61, 1–15. doi: 10.1109/TGRS.2023.3322787

Yuan, X., Li, J., Wang, W., Zhou, X., Li, N., and Yu, C. (2025). Improved YOLOv9 for underwater side scan sonar target detection. *Comput. J.* 68, 591–604. doi: 10.1093/comjnl/bxae134

Yung, N. D. T., Wong, W. K., Juwono, F. H., and Sim, Z. A. (2022). "Safety helmet detection using deep learning: implementation and comparative study using YOLOv5, YOLOv6, and YOLOv7", in: *2022 International Conference on Green Energy, Computing and Sustainable Technology (GECOST)* (Miri Sarawak: IEEE), 164–170. doi: 10.1109/GECOST55694.2022.10010490

Zhang, X. (2024). An efficient method for the simulation of multireceiver SAS raw signal. *Multimedia Tools Appl.* 83, 37351–37368. doi: 10.1007/s11042-023-16992-5

Zhang, X., Chen, X., and Qu, W. (2017a). "Influence of the stop-and-hop assumption on synthetic aperture sonar imagery", in: *2017 IEEE 17th International Conference on Communication Technology (ICCT)* (Chengdu: IEEE), 1601–1607. doi: 10.1109/ICCT.2017.8359901

Zhang, X., Dai, X., and Fang, B. (2019a). A range-doppler imaging method for the multireceiver synthetic aperture sonar. *Geomatics Inf. Sci. Wuhan Univ.* 44, 1667–1673. doi: 10.13203/j.whugis20180076

Zhang, X., Dai, X., and Yang, B. (2018a). Fast imaging algorithm for the multiple receiver synthetic aperture sonars. *IET Radar Sonar Navig.* 12, 1276–1284. doi: 10.1049/iet-rsn.2018.5040

Zhang, X., Huang, H., Ying, W., Wang, H., and Xiao, J. (2017b). An indirect range-doppler algorithm for multireceiver synthetic aperture sonar based on lagrange inversion theorem. *IEEE Trans. Geosci. Remote Sens.* 55, 3572–3587. doi: 10.1109/TGRS.2017.2676339

Zhang, Y., Li, S., Yao, T., Wang, S., Zhang, L., and Yan, L. (2025). "A drone detection network based on multi-scale feature fusion and content-guided attention mechanism", in: *2025 8th International Conference on Artificial Intelligence and Big Data (ICAIBD)* (Chengdu: IEEE), 219–224. doi: 10.1109/ICAIBD64986.2025.11082087

Zhang, X., Liu, Y., Liu, Y., and Deng, X. (2021). "Doppler centroid estimation for multireceiver SAS", in: *2021 13th International Conference on Communication Software and Networks (ICCSN)* (Chongqing: IEEE), 109–112. doi: 10.1109/ICCSN52437.2021.9463593

Zhang, T., Luo, B., Sharda, A., and Wang, G. (2022). Dynamic label assignment for object detection by combining predicted IoUs and anchor IoUs. *J. Imaging.* 8, 193. doi: 10.3390/jimaging8070193

Zhang, X., Tan, C., and Ying, W. (2019b). An imaging algorithm for multireceiver synthetic aperture sonar. *Remote Sens.*. 11, 672. doi: 10.3390/rs11060672

Zhang, X., Tang, J., and Zhong, H. (2014a). Multireceiver correction for the chirp scaling algorithm in synthetic aperture sonar. *IEEE J. Oceanic Eng.* 39, 472–481. doi: 10.1109/JOE.2013.2251809

Zhang, X., Tang, J., Zhong, H., and Zhang, S. (2014b). Wavenumber-domain imaging algorithm for wide-beam multi-receiver synthetic aperture sonar. *J. Harbin Eng. Univ.* 35, 93–101. doi: 10.3969/j.issn.1006-7043.201211060

Zhang, X., and Yang, P. (2022a). Back projection algorithm for multi-receiver synthetic aperture sonar based on two interpolators. *J. Mar. Sci. Eng.*. 10, 718. doi: 10.3390/jmse10060718

Zhang, X., Yang, P., and Cao, D. (2024). Synthetic aperture image enhancement with near-coinciding nonuniform sampling case. *Comput. Electr. Eng.* 120, 109818. doi: 10.1016/j.compeleceng.2024.109818

Zhang, X., Yang, P., and Dai, X. (2019). Focusing multireceiver SAS data based on the fourth-order legendre expansion. *Circuits Syst. Signal Process.* 38, 2607–2629. doi: 10.1007/s00034-018-0982-6

Zhang, X., Yang, P., Feng, X., and Sun, H. (2022b). Efficient imaging method for multireceiver SAS. *IET Radar Sonar Navig.* 16, 1470–1483. doi: 10.1049/rsn2.12274

Zhang, X., Yang, P., Huang, P., Sun, H., and Ying, W. (2022). Wide-bandwidth signal-based multireceiver SAS imagery using extended chirp scaling algorithm. *IET Radar Sonar Navig.* 16 (3), 531–541. doi: 10.1049/rsn2.12200

Zhang, X., Yang, P., and Sun, H. (2022c). Frequency-domain multireceiver synthetic aperture sonar imagery with chebyshev polynomials. *Electron. Lett.* 58, 995–998. doi: 10.1049/ell2.12513

Zhang, X., Yang, P., and Sun, H. (2023a). An omega-k algorithm for multireceiver synthetic aperture sonar. *Electron. Lett.* 59, e12859. doi: 10.1049/ell2.12859

Zhang, X., Yang, P., and Wang, J. (2026a). A quartic nonlinear chirp scaling algorithm for high-resolution wideband multireceiver SAS system. *Signal Process.* 250, 110815. doi: 10.1016/j.sigpro.2026.110815

Zhang, X., Yang, P., and Wang, J. (2026b). Precise imagery of multireceiver synthetic aperture sonar data based on range migration algorithm. *Signal Process.* 245, 110596. doi: 10.1016/j.sigpro.2026.110596

Zhang, X., Yang, P., Wang, J., and Zhu, J. (2026). Focus improvement of multireceiver SAS based on range-doppler algorithm. *IEEE Trans. Instrum. Meas.* 75, 1–14. doi: 10.1109/TIM.2026.3664370

Zhang, X., Yang, P., and Zhou, M. (2023b). Multireceiver SAS imagery with generalized PCA. *IEEE Geosci. Remote Sens. Lett.* 20, 1–5. doi: 10.1109/LGRS.2023.3286180

Zhang, X., and Ying, W. (2022d). Influence of the element beam pattern on synthetic aperture sonar imaging. *Geomatics Inf. Sci. Wuhan Univ.* 47, 133–140. doi: 10.13203/j.whugis20190148

Zhang, X., Ying, W., and Yang, B. (2018b). Parameter estimation for class A modeled ocean ambient noise. *J. Eng. Technol. Sci.* 50, 330–345. doi: 10.5614/j.eng.technol.sci.2018.50.3.2

Zhang, R., Zhu, W., Li, Z., Zhang, B., and Chen, B. (2024). Re-Net: Multibranch network with structural reparameterization for landslide detection in optical imagery. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 17, 2828–2837. doi: 10.1109/JSTARS.2023.3344720

Zhang, F., and Zhuge, J. (2024). "Modeling and solving of seafloor terrain detection based on multibeam bathymetry", in: *2024 IEEE 3rd International Conference on Electrical Engineering, Big Data and Algorithms (EEBDA)* (IEEE), 235–241. doi: 10.1109/EEBDA60612.2024.10485903

Zhao, L., and Li, S. (2020). Object detection algorithm based on improved YOLOv3. *Electronics*. 9, 537. doi: 10.3390/electronics9030537

Zhao, S., Luo, Y., Zhang, T., Guo, W., and Zhang, Z. (2023). A domain specific knowledge extraction transformer method for multisource satellite-borne SAR images ship detection. *ISPRS J. Photogramm. Remote Sens.* 198, 16–29. doi: 10.1016/j.isprsjprs.2023.02.011

Zhao, S., Zhang, Y., Luo, Y., Kang, Y., and Wang, H. (2024). Dynamically self-training open set domain adaptation classification method for heterogeneous SAR image. *IEEE Geosci. Remote Sens. Lett.* 21, 1–5. doi: 10.1109/LGRS.2024.3360006

Zhao, H., Zhang, H., and Zhao, Y. (2023).YOLOv7-Sea: Object detection of maritime UAV images based on improved YOLOv7. Available online at: https://openaccess.thecvf.com/content/WACV2023W/MaCVi/html/Zhao_YOLOv7-Sea_Object_Detection_of_Maritime_UAV_Images_Based_on_Improved_WACVW_2023_paper.html (Accessed May 7, 2026).

Zhao, X., Zhao, S., Luo, Y., Lv, B., and Zhang, Z. (2025). Generalizing SAR object detection: A unified framework for cross-source scenarios. *IEEE Geosci. Remote Sens. Lett.* 22, 1–5. doi: 10.1109/LGRS.2025.3575173

Zheng, K., Liang, H., Zhao, H., Chen, Z., Xie, G., Li, L., et al (2024). Application and analysis of the MFF-YOLOv7 model in underwater sonar image target detection. *J. Mar. Sci. Eng.*. 12, 2326. doi: 10.3390/jmse12122326

Zheng, Z., Wang, P., Liu, W., Li, J., Ye, R., and Ren, D. (2020). "Distance-IoU loss: faster and better learning for bounding box regression", in: *Proceedings of the AAAI Conference on Artificial Intelligence* (AAAI Press), 34, 12993–13000. doi: 10.1609/aaai.v34i07.6999

Zheng, J., Zhao, S., Xu, Z., Zhang, L., and Liu, J. (2023). Anchor boxes adaptive optimization algorithm for maritime object detection in video surveillance. *Front. Mar. Sci.* 10. doi: 10.3389/fmars.2023.1290931

Zhou, W., and Xu, G. (2024). "SAR image aircraft target detection based on multi-auxiliary head supervision network", in: *2024 6th International Conference on Electronic Engineering and Informatics (EEI)* (Chongqing: IEEE), 536–540. doi: 10.1109/EEI63073.2024.10696644